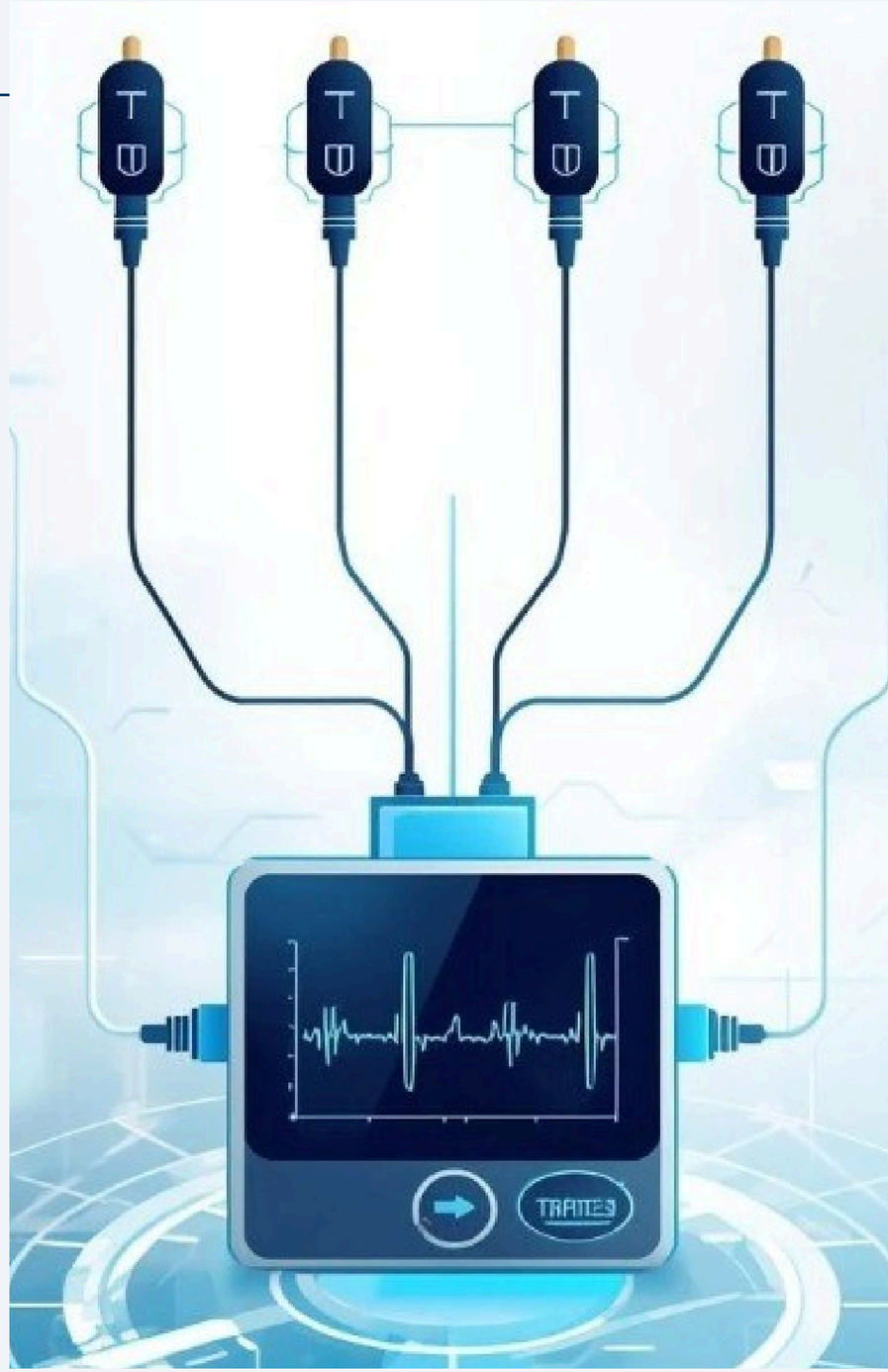
Group ID: MON-04

Simultaneous Measurement of IPG and ECG using dry Electrodes

Milestone 2

Team Members- Yashvardhan, Krunal, Sudhindra, Vinay, Jahnvi



Milestone 1 Summary: Work Completed and Feedback Addressed

Progress

- Literature review to comprehensively understand the problem statement
- Identified potential ways of implementing, considered feasibility & reliability for each, and decided best method
- Did preliminary system design, including the circuit diagrams, GUI and component identification

Feedback

The work completed was well-received and found to cover most aspects. We were advised to further validate that decisions regarding the number and placement of electrodes are well-justified. Two areas were identified for additional work:

- Incorporating the necessary interfacing components for the external oscillator
- Researching appropriate testing methods, as direct testing on humans may not be safe.

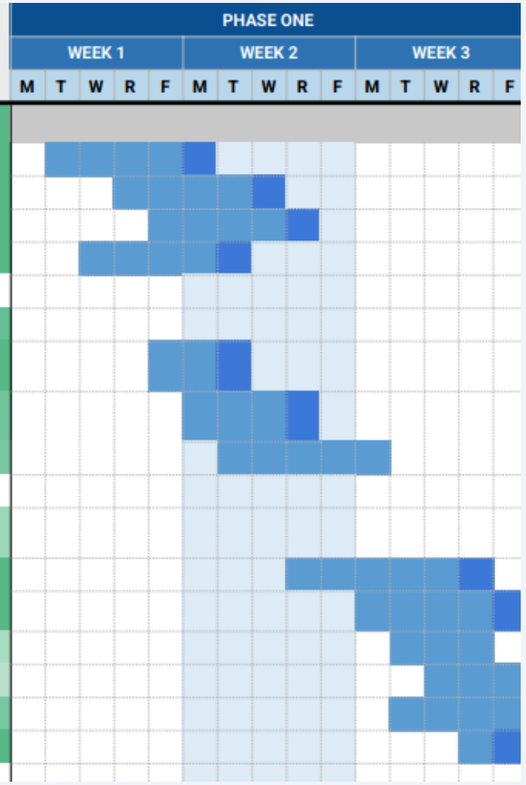
Changes made

- Finalized number of electrodes to be used: 5 electrodes configuration for simultaneous ECG and dual-IPG acquisition.
- Included an external oscillator along with its required interfacing components to ensure accurate and stable timing for analog front-end operation.
- Selected a single implementation approach: chip-based architecture using dedicated biomedical AFE ICs instead of discrete analog design.
- Adopted thorax impedance simulator as a testing tool for validating impedance measurement accuracy before human testing.

Gantt chart

Gantt chart consisting of the 3 weeks covered till now

WBS NUMBER	TASK TITLE	TASK OWNER	START DATE	DUE DATE	DURATION	% OF TASK COMPLETE
1	**Literature Review**		1/15/26	1/28/26	13	100%
1.1	Search for research papers	Krunal	1/15/26	1/18/26	3	100%
1.1.1	Paper Reading	All	1/18/26	1/28/26	10	100%
1.2	Documentation of concepts	Vinay	1/28/26	2/7/26	9	100%
1.3	Rough circuit diagram and specs	Jhanvi	2/7/26	2/10/26	3	100%
2	**BoM and Initial Design Review**		2/10/26	2/16/26		90%
2.1	Scope and Goal Setting	Yashvardhan, Vinay	2/10/26	2/13/26	4	100%
2.2	**Budget**	**Sudhindra, Jhanvi**	2/13/26	2/14/26	2	100%
2.3	Communication to vendors	Jhanvi	2/14/26	2/16/26	2	80%
3	**Circuit Verification and Implementation**		2/17/26	2/28/26		60%
3.1	Status and Tracking	Yashvardhan	2/17/26	2/20/26	3	100%
3.2	**PCB schematic design**	**Sudhindra**	2/17/26	2/22/26	5	100%
3.2.1	Monitoring	Jhanvi	2/20/26	2/25/26	5	50%
3.2.2	SPI working	Jhanvi	2/23/26	2/28/26	5	40%
3.3	Documentation	Vinay	2/17/26	2/28/26	11	80%
3.3.1	Chart Updates	Krunal	2/20/26	2/28/26	8	100%



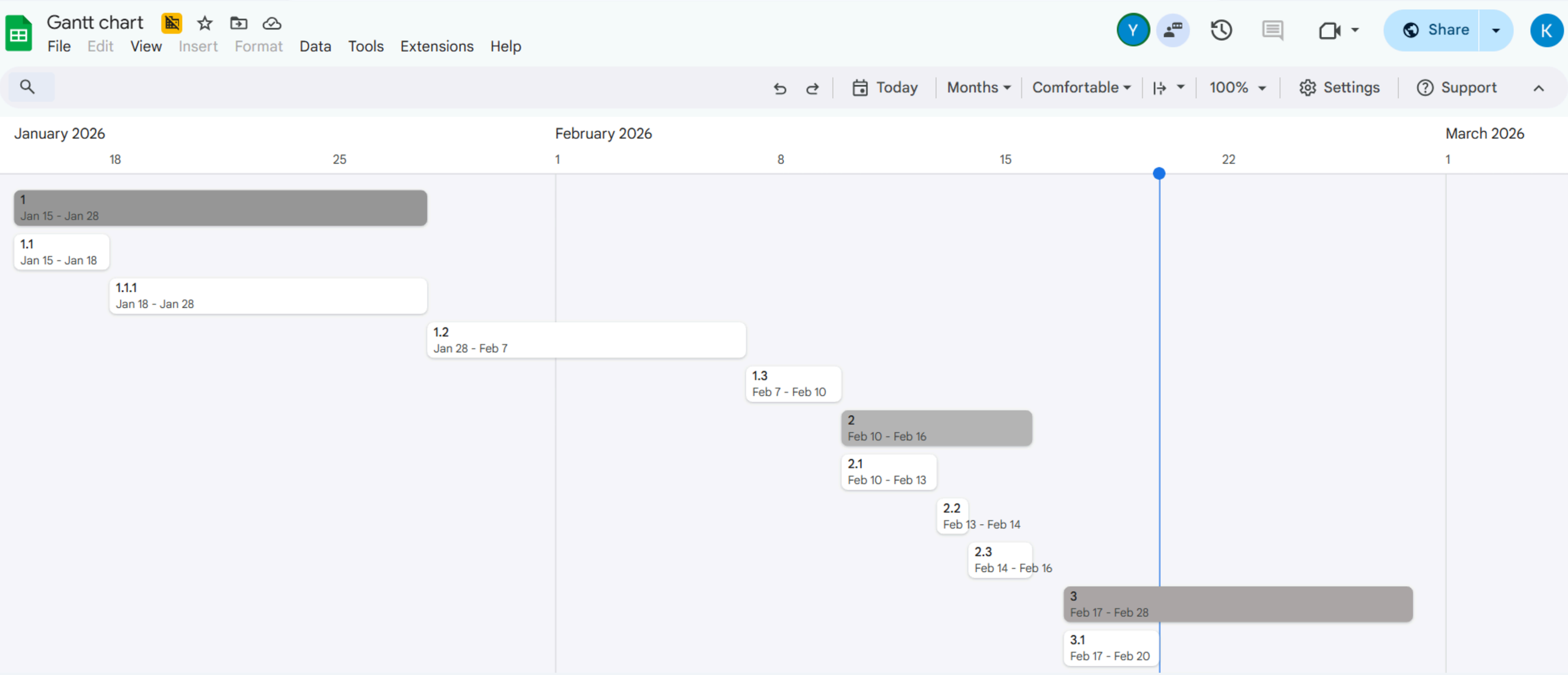
4	**Troubleshooting**		3/1/26	3/10/26	
4.1	ECG and IPG processing	Vinay	3/1/26	3/3/26	2
4.2	BMA	Sudhindra, Yashvardhan	3/3/26	3/6/26	3
4.3	Inter-system communication	Jhanvi	3/4/26	3/8/26	4
4.4	Waveform-measurement	Krunal, Yashvardhan	3/6/26	3/10/26	4
5	**Prototype-1 Design Start**		3/11/26	3/25/26	
5.1	PCB confirmation	Yashvardhan	3/11/26	3/13/26	2
5.2	Product design	Sudhindra, Vinay	3/14/26	3/17/26	3
5.3	Whole testing	Jhanvi, Vinay	3/18/26	3/21/26	3
5.4	New and reliable test methods	Krunal	3/22/26	3/25/26	3
6	**Breadboard Analysis & Proto-1 Design**		3/26/26	4/5/26	
6.1	PCB confirmation	Yashvardhan	3/26/26	3/28/26	2
6.2	Product design	Sudhindra, Vinay	3/29/26	4/1/26	3
6.3	Whole testing	Jhanvi, Vinay	4/2/26	4/4/26	2
6.4	New and reliable test methods	Krunal	**4/4/26**	**4/5/26**	**1**
7	**PCB Final Verification**		4/6/26	4/15/26	
7.1	PCB confirmation	Yashvardhan	4/6/26	4/8/26	2
7.2	Product design	Sudhindra, Vinay	4/9/26	4/11/26	2
7.3	Whole testing	Jhanvi, Vinay	**4/12/26**	**4/14/26**	**2**
7.4	New and reliable test methods	Krunal	4/14/26	4/15/26	1
8	**Final Results**		4/16/26	4/22/26	
8.1	PCB confirmation	Yashvardhan	4/16/26	4/17/26	1
8.2	Product design	Sudhindra, Vinay	**4/18/26**	**4/19/26**	**1**
8.3	Whole testing	Jhanvi, Vinay	4/20/26	4/21/26	1
8.4	New and reliable test methods	Krunal	4/21/26	4/22/26	1
9	**Final Presentation and Report**		4/23/26	4/30/26	
9.1	PCB confirmation	Yashvardhan	4/23/26	4/24/26	1
9.2	Product design	Sudhindra, Vinay	4/25/26	4/26/26	1
9.3	Whole testing	Jhanvi, Vinay	4/27/26	4/28/26	1

Plan for our project:
Please note that the specifics are still rough after week 5

Link to gantt chart and timeline:

https://docs.google.com/spreadsheets/d/1mrR8jPvj8YX_cm8KRvoaAJXosBHN-fHIcQ4bGQeruhQ/edit?gid=1115838130#gid=1115838130

Timeline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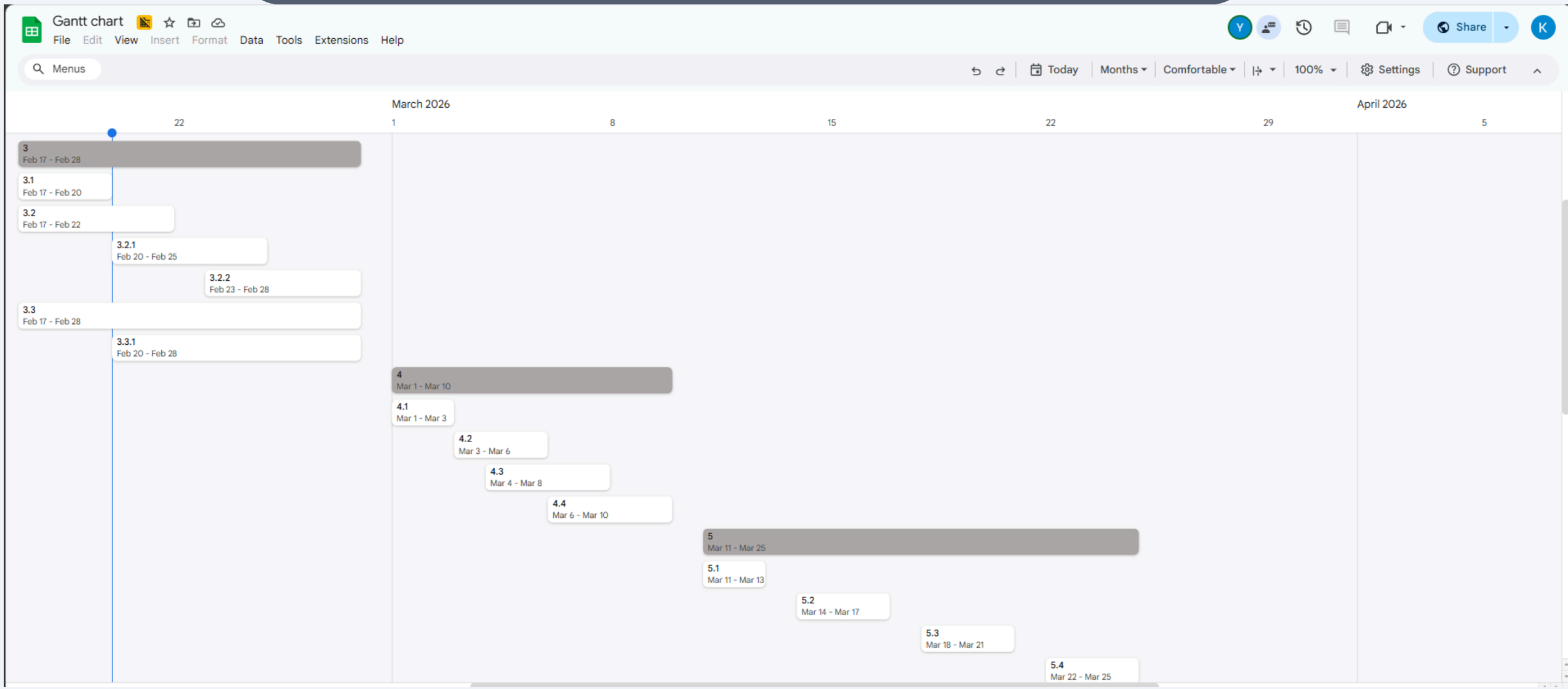
Timeline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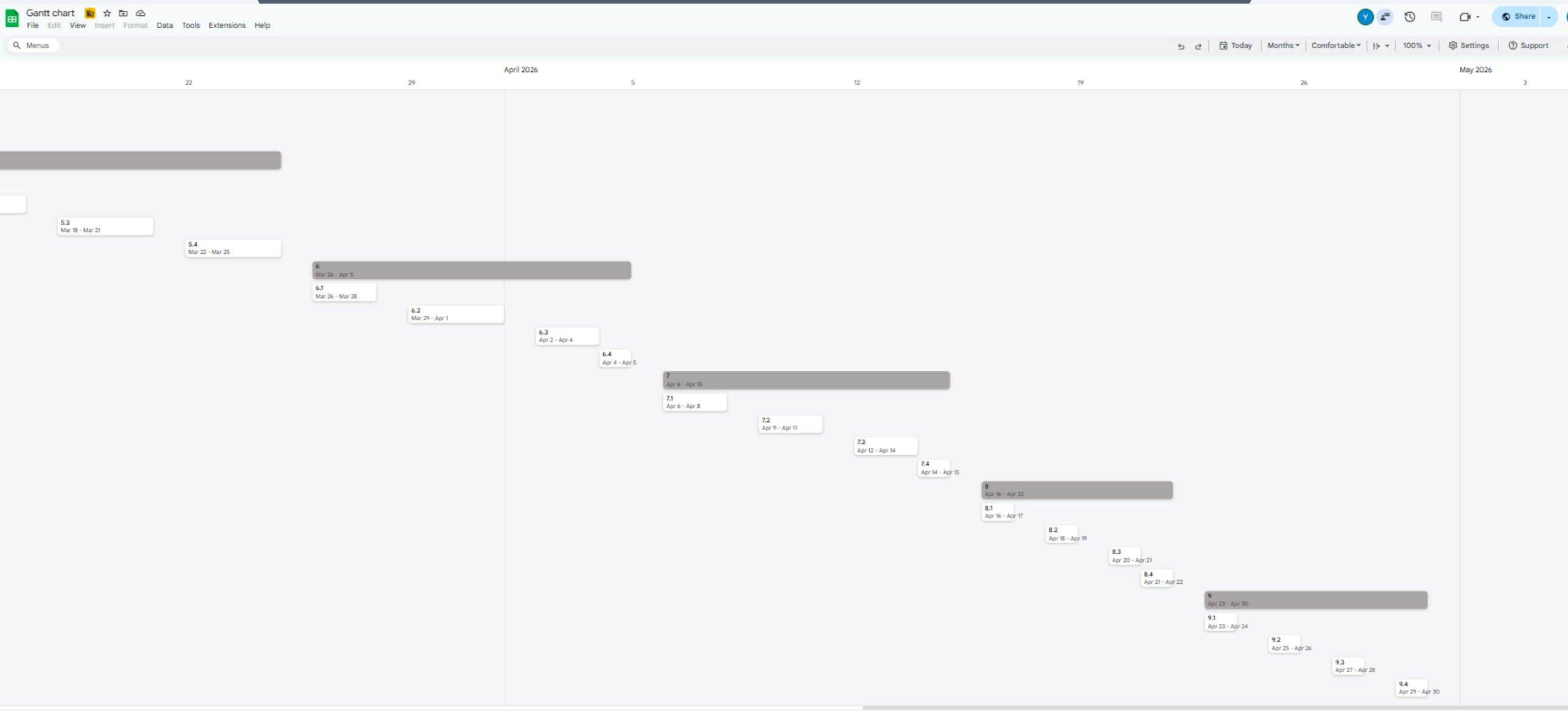
Timeline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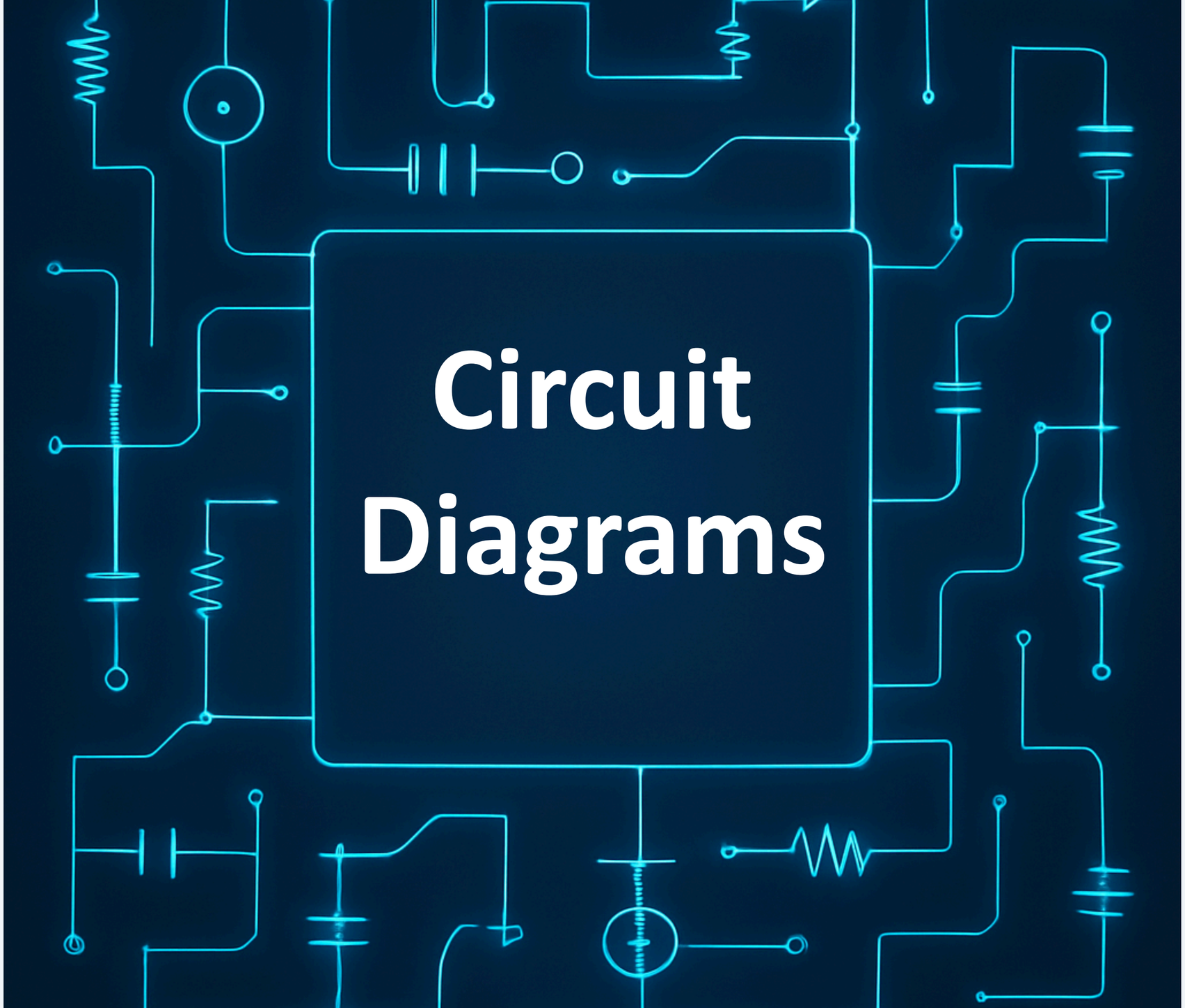
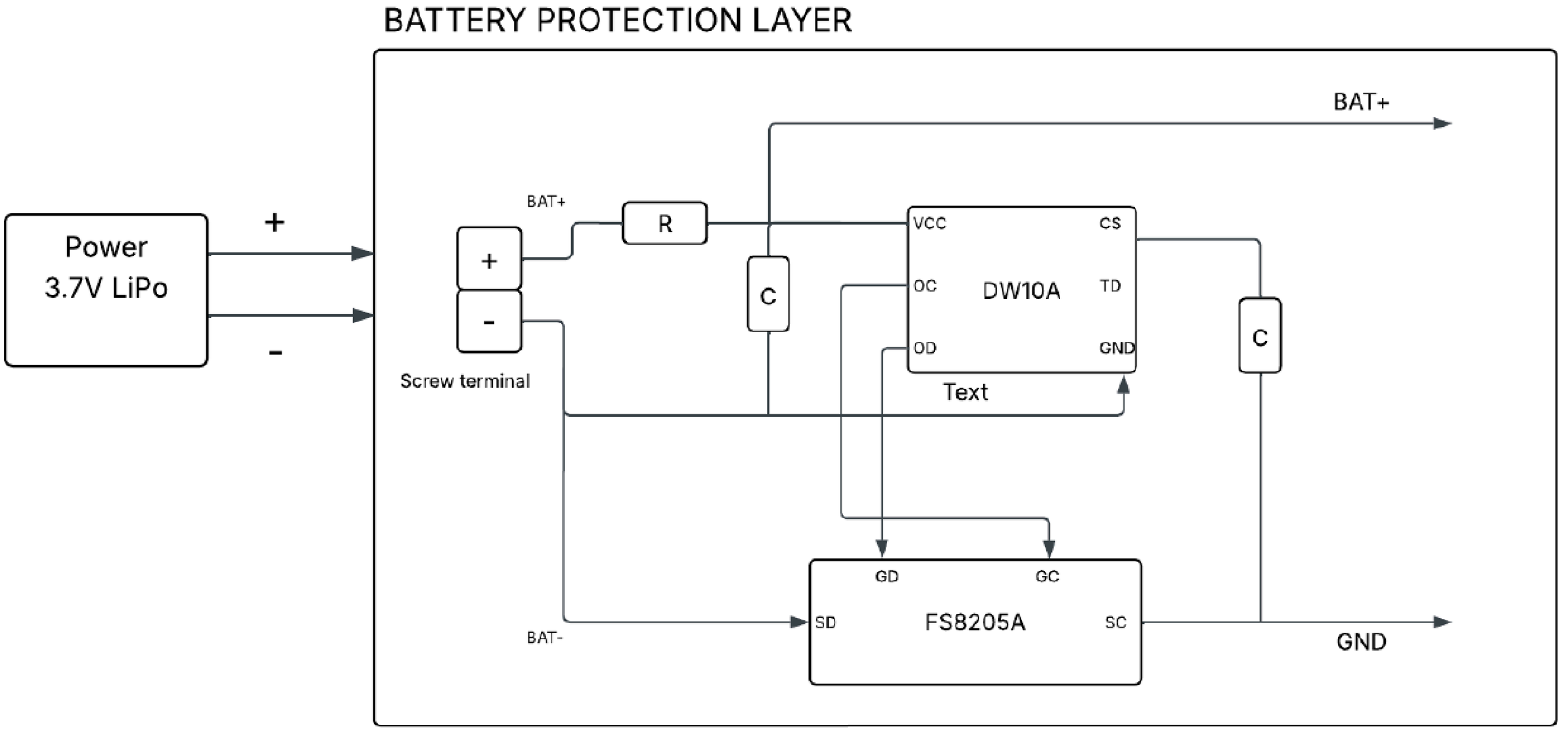
Apr 25 - Apr 26
9.3
Apr 27 - Apr 28
9.4
Apr 29 - Apr 30

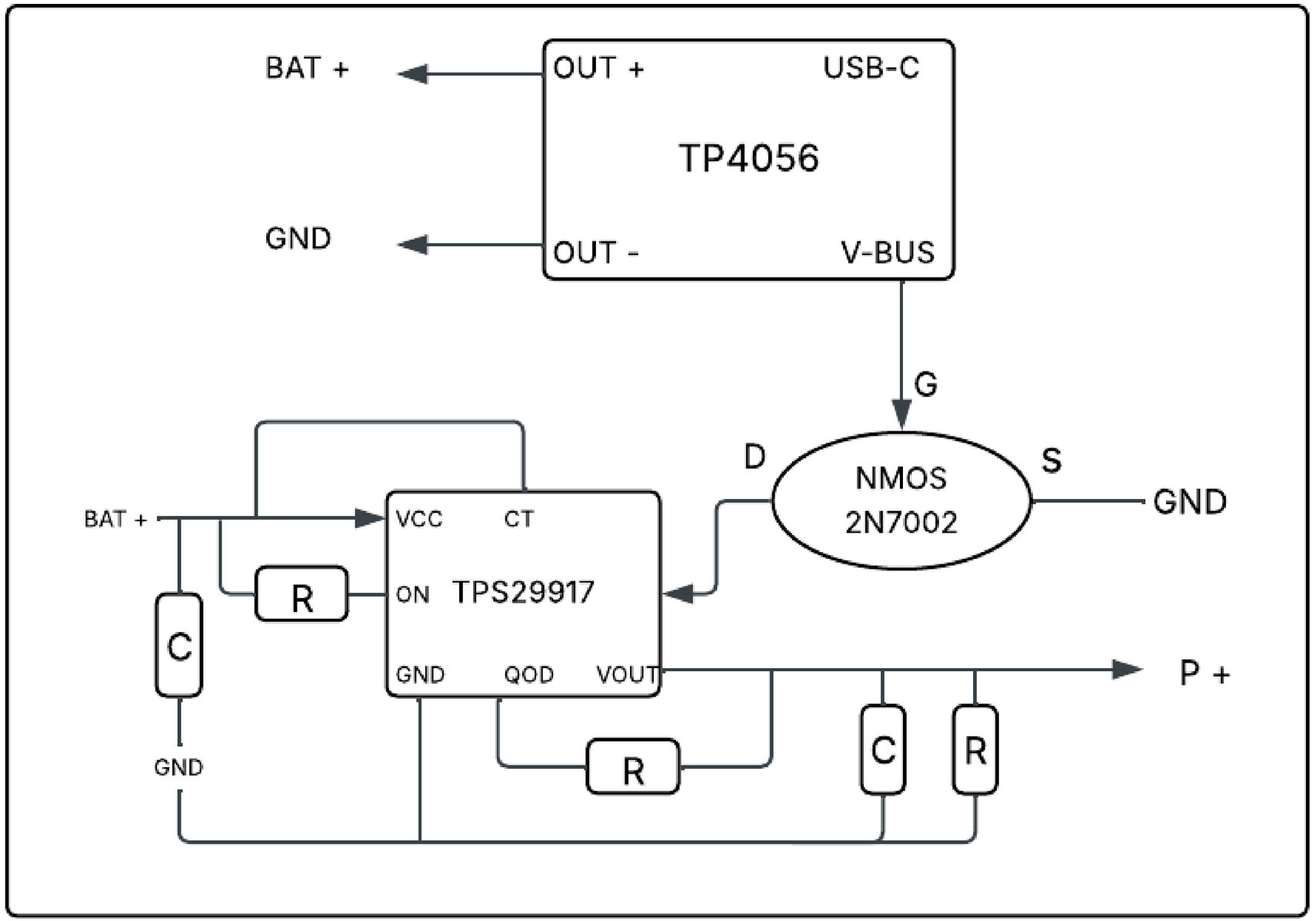
Changes made against gantt chart

Hardware Decisions	Testing Protocol	Accuracy Concerns	Timeline Impact
Our initial design featured four electrodes per chip integrated into a custom PCB. However, following a Bill of Materials (BOM) review with the research/teaching assistants and Maheshwar Sir, we pivoted away from direct chip integration due to the technical difficulties of soldering BGA (Ball Grid Array) packages. Per their advice, we are now utilizing breakout boards. This modification reduces the available pins to three electrodes per chip (including the RLD drive); consequently, our dual-chip configuration will now utilize a total of five electrodes.	While we originally intended to conduct human subject testing, we have shifted to using a proxy (e.g., a signal generator or phantom) to address safety concerns. (as suggested by Himanshu Sir)	Due to simultaneous current injection and voltage measurement, the quality of the measured IPG signal will also degrade, as PC Pandey Sir pointed out to us.	We are currently behind the original Gantt chart schedule. The "System-on-Chip" testing phase has been delayed due to the late delivery of the integrated circuits, which are critical components of our system.

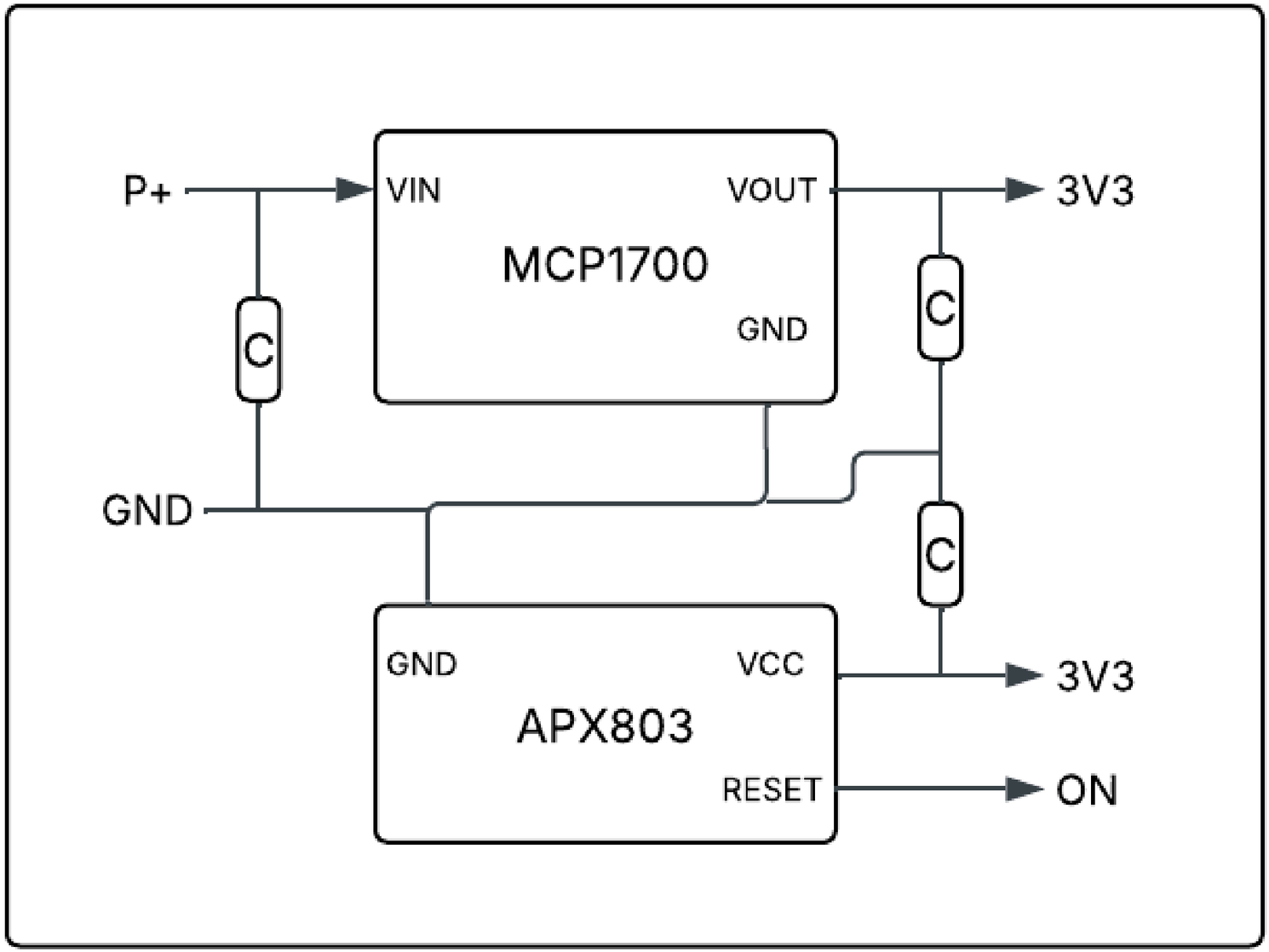

Circuit
Diagrams


BATTERY PROTECTION LAYER
Power
3.7V LiPo
+
-
BAT+
R
+
-
Screw terminal
C
VCC
CS
OC
DW10A
TD
OD
GND
Text
C
BAT+
BAT-
GD
GC
SD
FS8205A
SC
GND

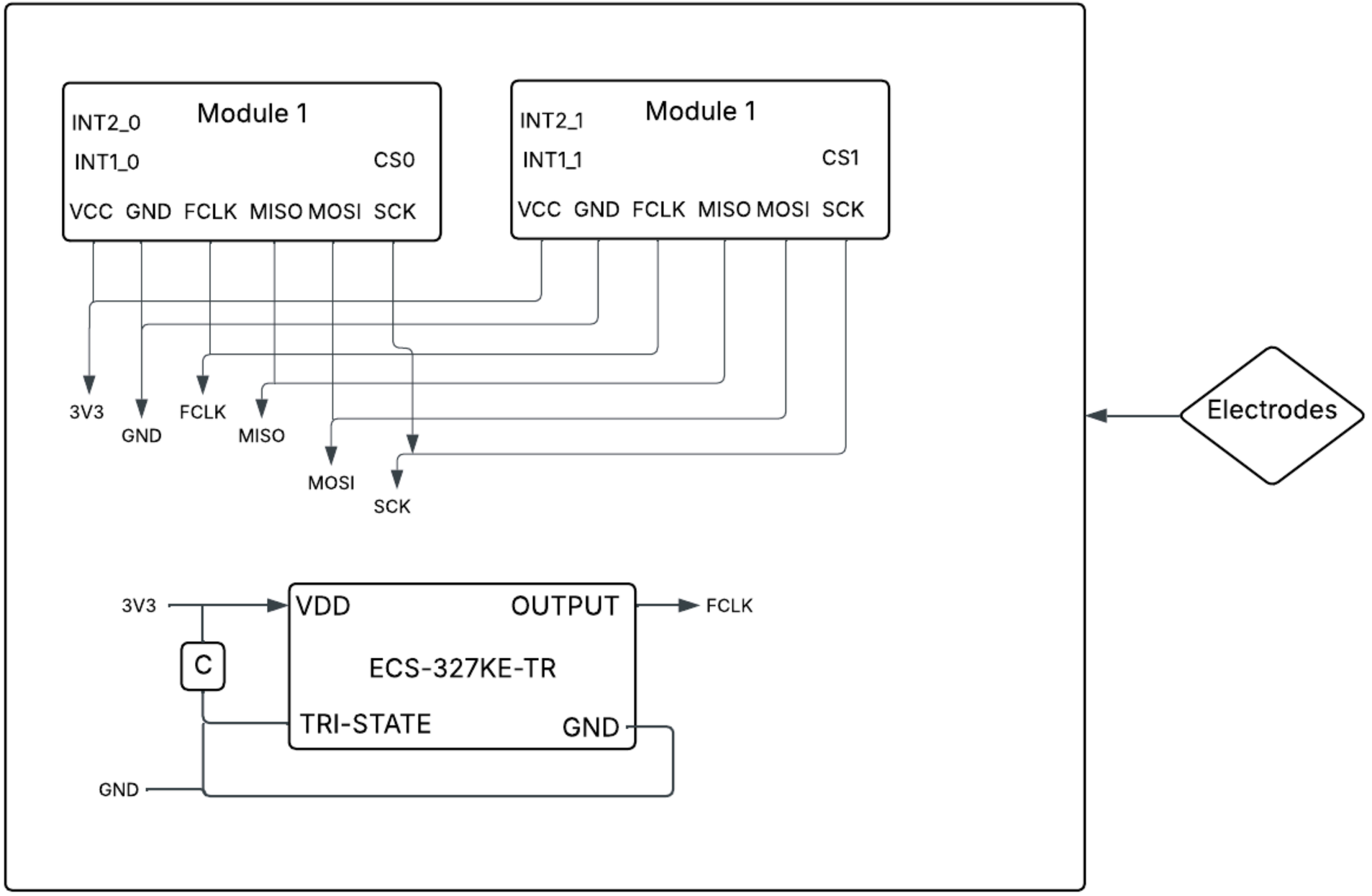
BATTERY CHARGING LAYER



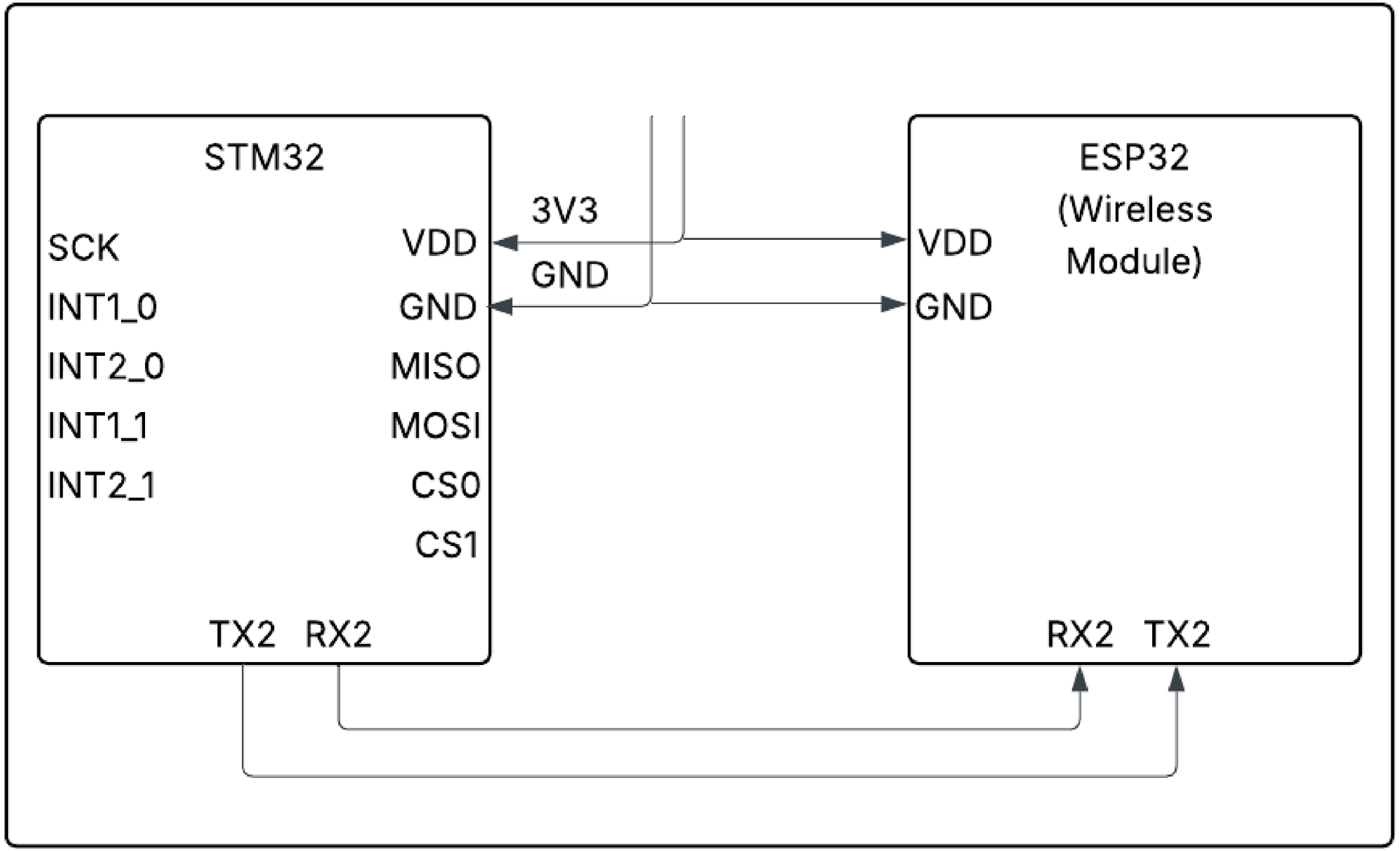
VOLTAGE REGULATION LAYER



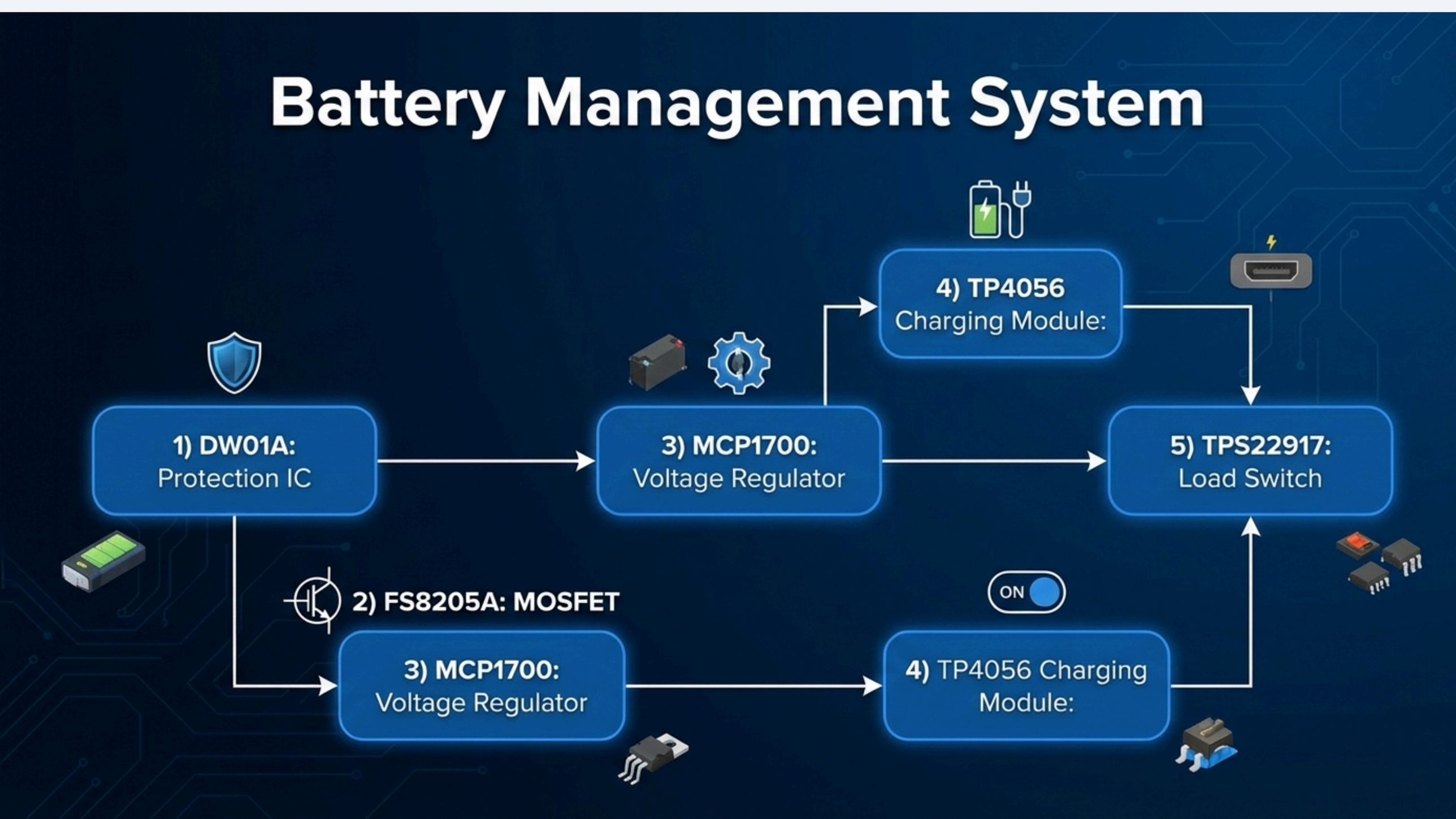
MAX30001 BREAKOUT BOARD

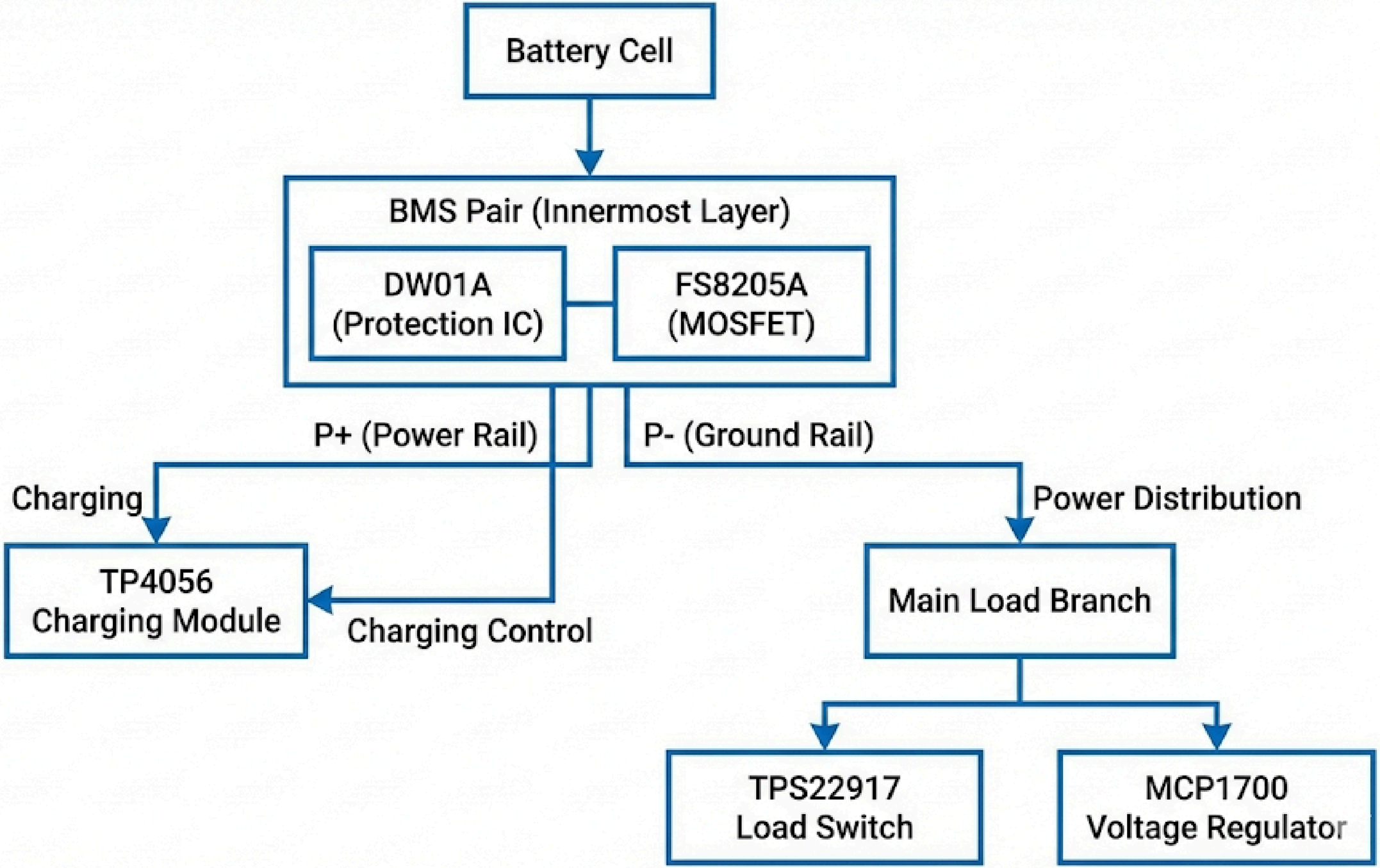


MCU & WIRELESS MODULE PINOUT

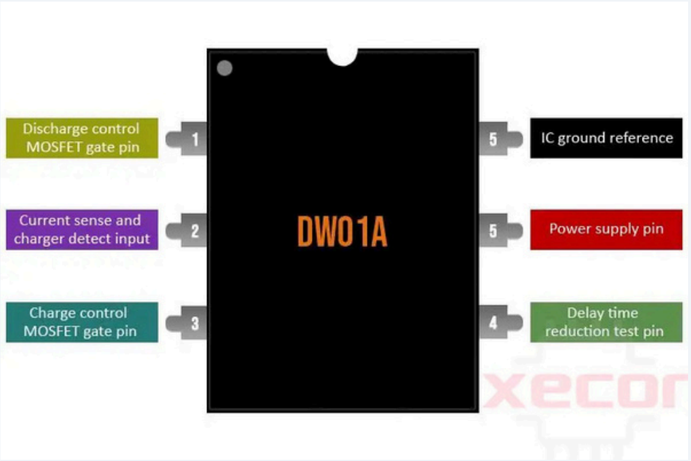


Battery Management System

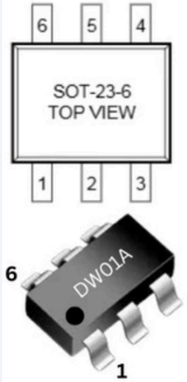



Battery Cell
BMS Pair (Innermost Layer)
DW01A
(Protection IC)
FS8205A
(MOSFET)
P+ (Power Rail)
P- (Ground Rail)
Charging
Power Distribution
TP4056
Charging Module
Charging Control
Main Load Branch
TPS22917
Load Switch
MCP1700
Voltage Regulator

1) DW01A:



DW01A Pinout



Pin No.	Symbol	Description
1	OD	MOSFET gate connection pin for discharge control
2	CS	Input pin for current sense, charger detect
3	OC	MOSFET gate connection pin for charge control
4	TD	Test pin for reduce delay time
5	VCC	Power supply, through a resistor (R1)
6	GND	Ground pin

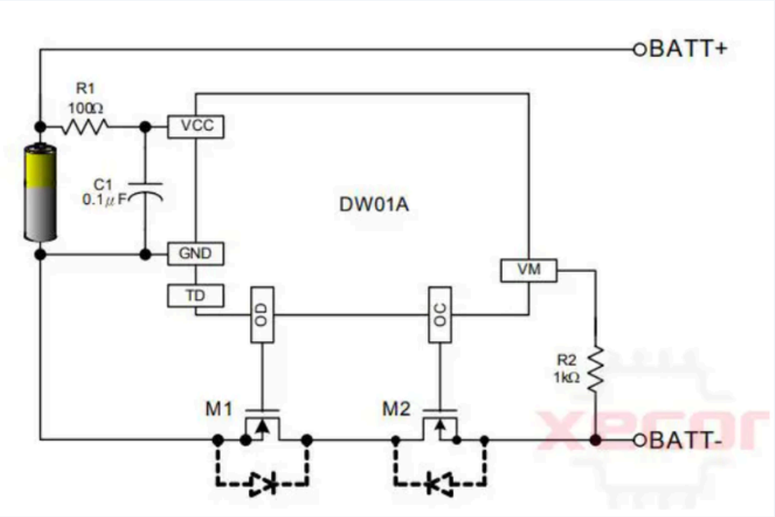


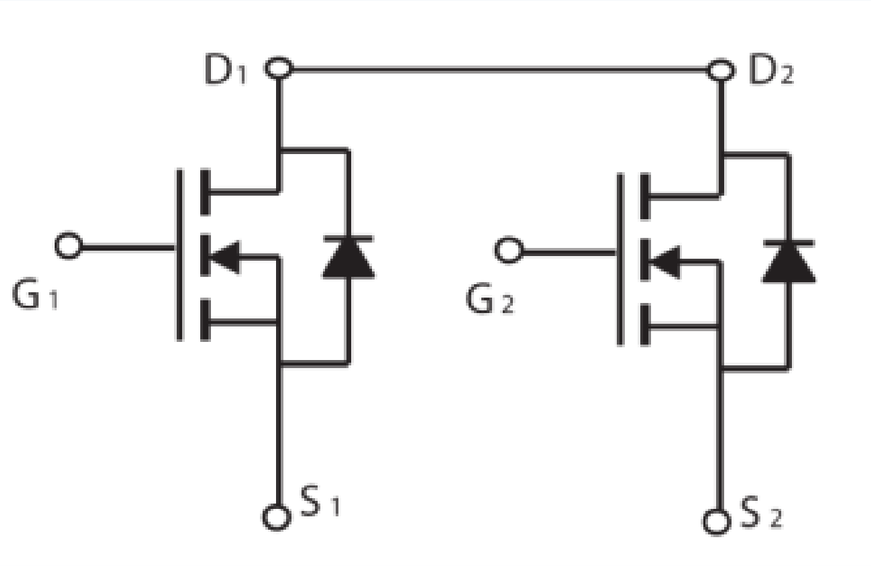
Fig 3: Application circuit for the DW01A

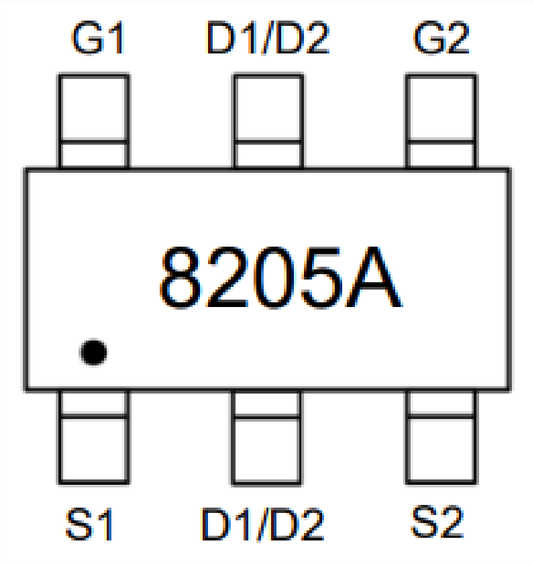
The DW01A is the control center for your Battery Management System (BMS). It continuously monitors the battery's voltage and current to decide when to open or close the MOSFET switches.

DW01A Pin Function Descriptions

PIN NAME	FUNCTION	DESCRIPTION
OD (Over-Discharge)	Gate driver for Discharge MOSFET.	Turns MOSFET off if voltage drops below ~2.4V–3.0V to prevent damage.
OC (Over-Charge)	Gate driver for Charge MOSFET.	Turns MOSFET off if voltage exceeds safe limit to stop charging.
CS (Current Sense)	Monitors voltage drop to detect Overcurrent/Short Circuits.	Triggers protection at a specific limit; detects charger with negative voltage sensing.
VCC, GND	Power Supply and Ground.	Essential connections for device operation.

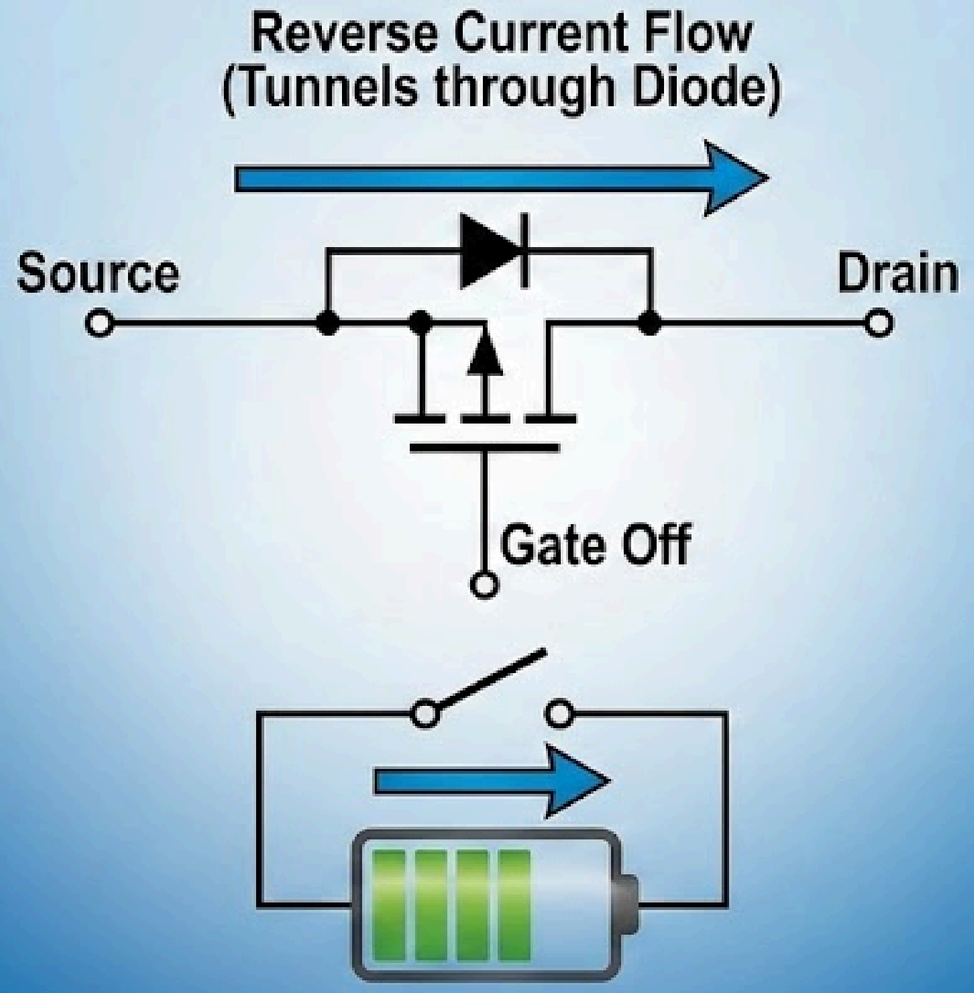
2) FS8205A:



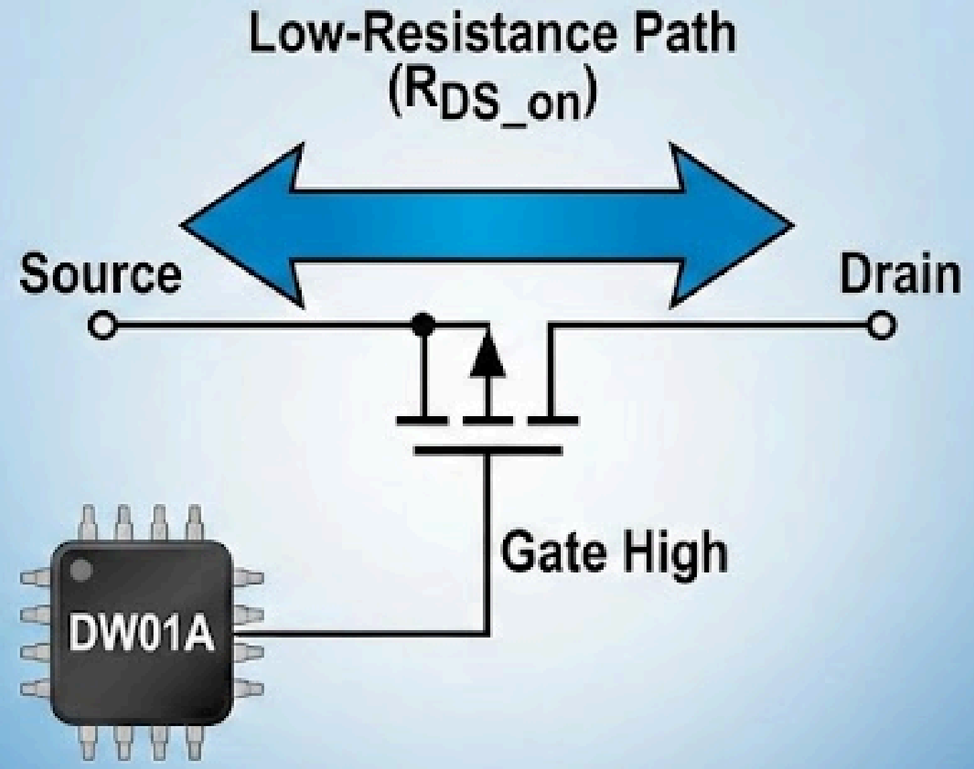


- Source of the "Discharge" MOSFET to the B- (Battery Negative).
- Source of the "Charge" MOSFET to the P- (the external Load/Charger negative terminal)
- Connect the OD pin (Pin 1) of the DW01A to the Gate of the first MOSFET in the FS8205A.
- Connect the OC pin (Pin 3) of the DW01A to the Gate of the second MOSFET in the FS8205A.

THE BODY DIODE

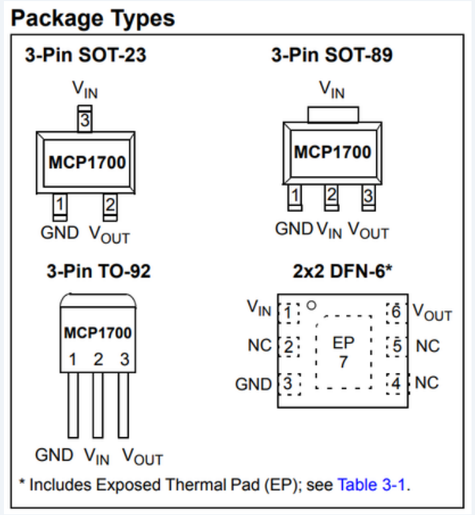


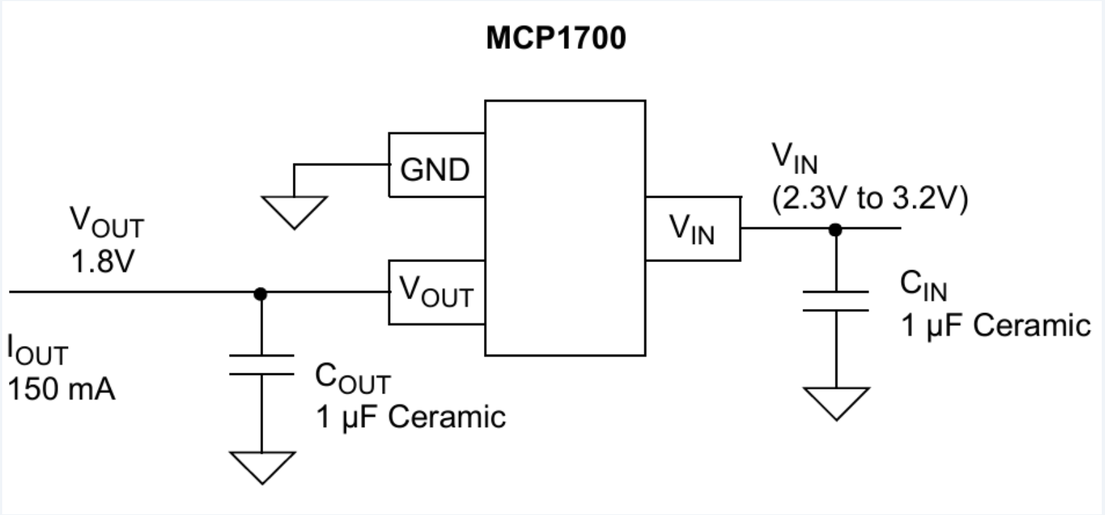
BILATERAL CHANNEL CONDUCTION

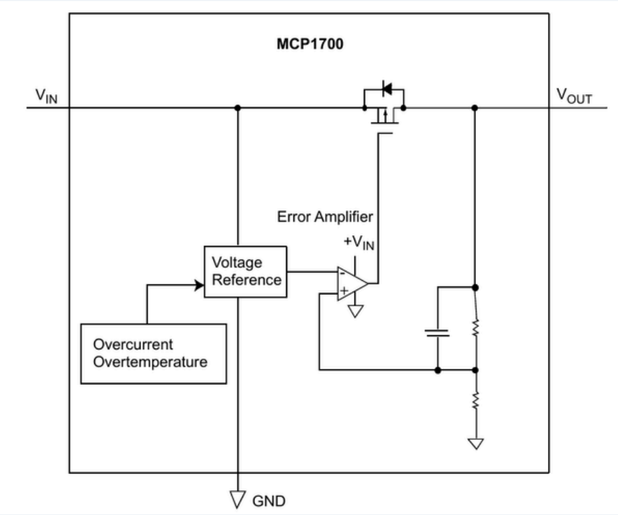


Current flows equally well in both directions when Gate is ON, bypassing the diode.

3) MCP1700:



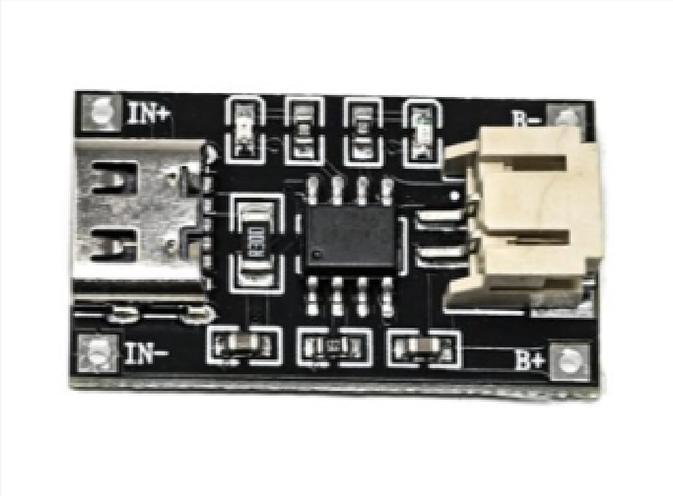


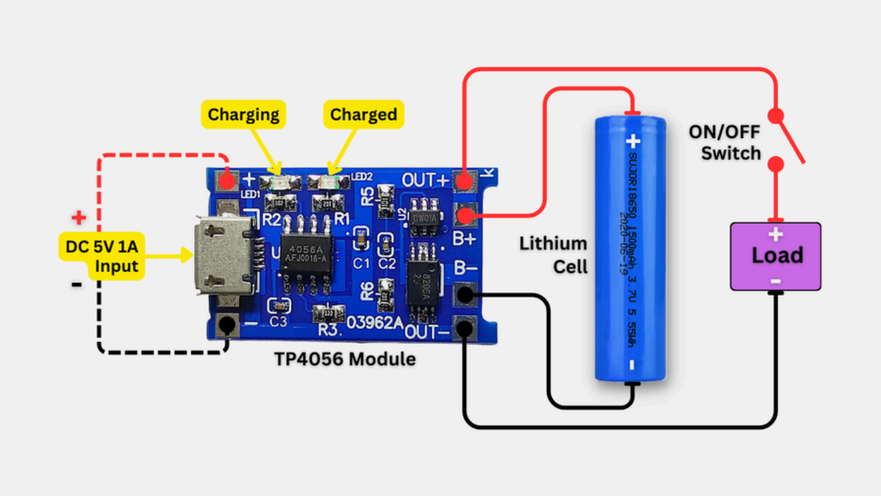


The MCP1700 is a CMOS Low Dropout (LDO) voltage regulator

- GND: Ground reference for the IC, from BMS P- Terminal
- V_OUT: Regulated output voltage (e.g., 3.3V), From TPS22917 VOUT
- V_IN: Unregulated input voltage from the battery, To Microcontroller/Sensors (e.g., ECG/Bio-impedance module)

4) TP4056 Charging module:





Charging Module (TP4056): Connect the BAT+ and BAT- pins of the TP4056 directly to the P+ and P- terminals the BMS.

Note: Do not connect the TP4056 directly to the battery cell; it must sit "outside" the BMS so the protection IC can cut off the charger if it malfunctions.

4) TPS22917:

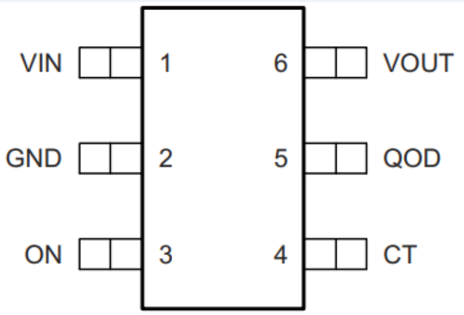
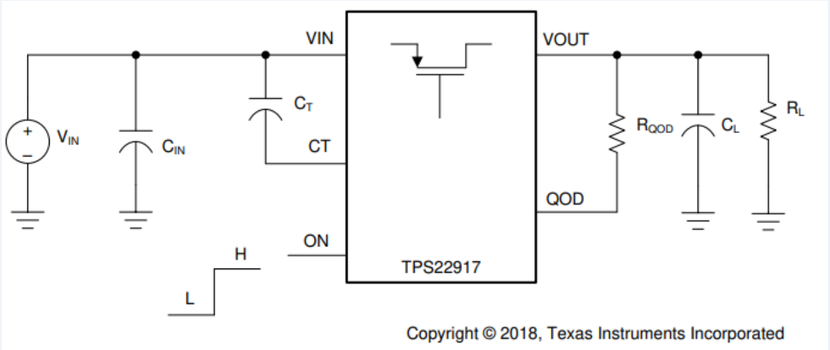


Figure 6-1. DBV Package 6-Pin SOT-23 Top View



Simplified Schematic

how to put it in the PCB

Implementing the "Disconnect while Charging" Logic

Input Power: Connect P+ (from BMS) to the VIN of the TPS22917.
Control Signal (ON Pin): The TPS22917 "ON" pin is active high. To turn the circuit OFF when a charger is plugged in, we need to "pull down" this pin when voltage is detected at the TP4056 VBUS/USB IN.
Simple Logic: We can use a small NPN transistor or a P-channel MOSFET. Connect the gate/base to the TP4056 input 5V. When 5V is present (charging), the transistor pulls the TPS22917 ON pin to Ground, opening the circuit.
Output to Regulator: Connect the VOUT of the TPS22917 to the VIN of the MCP1700 LDO.

Description of useful pins:

- V_OUT: Switched output voltage to the MCP1700 regulator.
- GND: Ground reference for the IC.
- ON: Active-high enable pin. High = V_{IN} connected to V_{OUT}; Low = Disconnected.
- V_{IN}: Input voltage from the P+ terminal of the BMS.
- CT: Charge Time control. Connect a capacitor here to set the "soft-start" (slew rate).
- QOD: Quick Output Discharge. Connect to V_{OUT} to drain the line fast when switched off

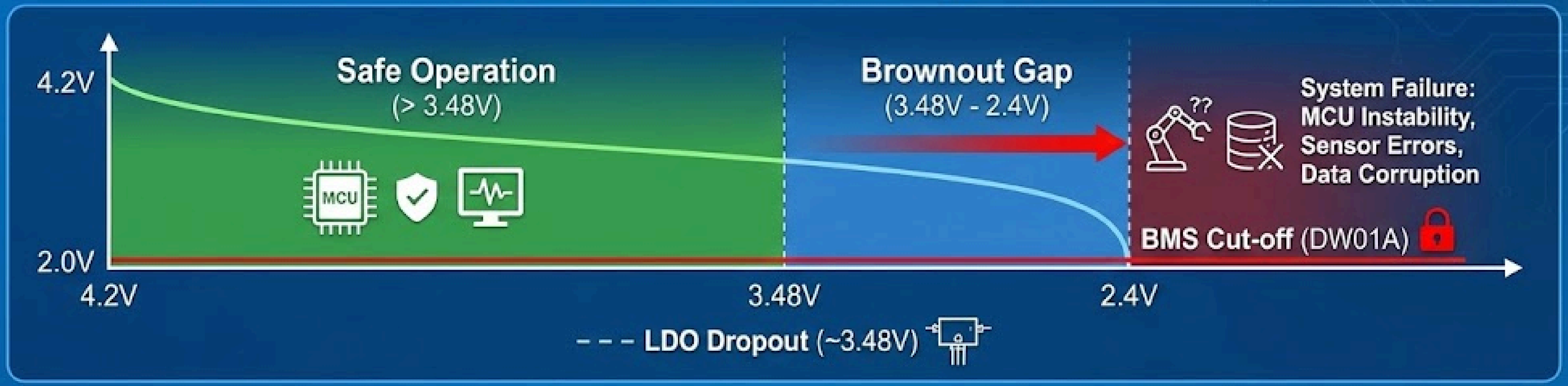
Battery Protection vs. System Stability: The "Brownout Gap"

- BMS Cut-off: The DW01A typically cuts power at 2.4V to prevent permanent battery damage.
- LDO Dropout: For a 3.3V output, the MCP1700 requires an input of at least ~3.48V due to its dropout voltage.
- System Failure: Between 3.4V and 2.4V, the regulator output drops, causing MCU instability, sensor errors, or data corruption in projects like your health monitors or robotic arms.

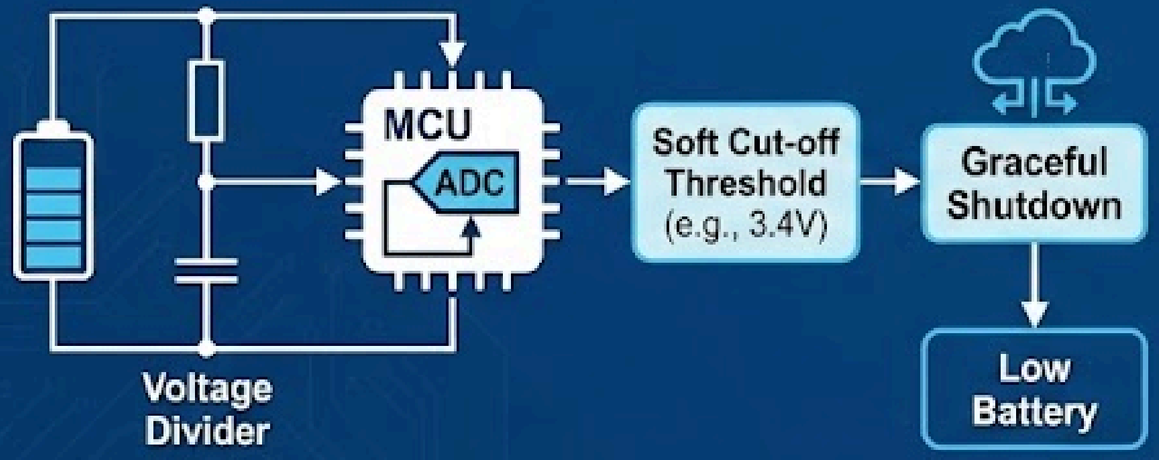
1) Software-Based Monitoring (Internal ADC)
Use a voltage divider to scale the battery voltage and read it via the MCU’s Analog-to-Digital Converter (ADC).
Program a "Soft Cut-off" threshold (e.g., 3.4V).
Allows the MCU to perform a graceful shutdown (saving data to EEPROM or sending a final "Low Battery" alert) before the voltage becomes unreliable.

2) Hardware Supervisor (External Reset IC)
Add a dedicated Voltage Supervisor IC (e.g., APX803) to monitor the 3.3V rail.
If the rail dips below a safe threshold (e.g., 3.0V), the supervisor pulls the TPS22917 "ON" pin or the MCU Reset pin low.
Provides instant, failsafe protection that does not rely on software logic, ensuring the system never runs in an undefined "glitch" state.

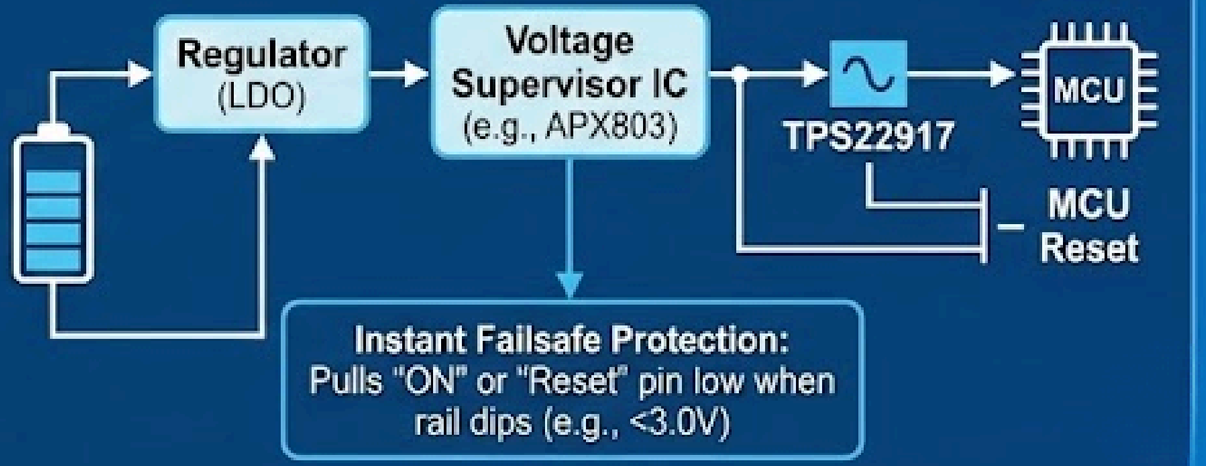
Battery Protection vs. System Stability: The "Brownout Gap"



1) Software-Based Monitoring (Internal ADC)



2) Hardware Supervisor (External Reset IC)

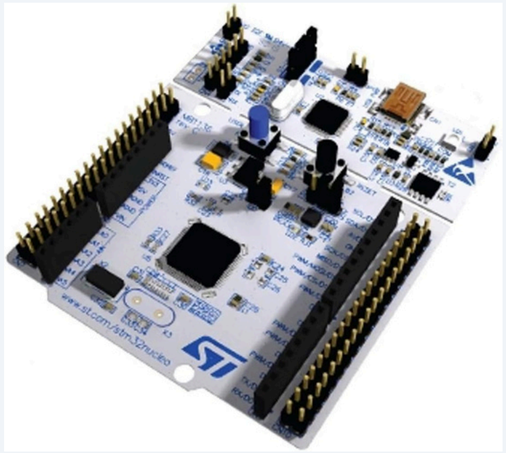


MAX30001 Breakout kit

The MAX30001 is a high-performance Analog Front-End (AFE) designed for clinical-grade ECG and bio-impedance (BioZ) measurements.

- VCC: Main analog supply pin. It is typically connected to 5V to power the internal analog circuitry.
- 3V3: Digital logic supply pin. It connects to a regulated 3.3V rail (for example, from an MCP1700) to match the MCU's logic levels.
- GND: Common ground reference used for both power return and signal reference.
- MISO (Master In Slave Out): SPI data output pin. It sends digitized ECG and BioZ data from the MAX30001 to the microcontroller.
- MOSI (Master Out Slave In): SPI data input pin. It receives configuration commands and control data from the microcontroller.
- SCK (Serial Clock): SPI clock signal. It synchronizes data transfer timing between the MAX30001 and the microcontroller.
- CS0 (Chip Select): Active-low SPI select pin. The microcontroller pulls this low to start communication with the MAX30001.
- INT1 / INT2: Programmable interrupt output pins. These notify the microcontroller about events such as lead-off detection, FIFO buffer full, or R-to-R heart rate detection.
- FCLK: External clock input pin. It allows a precise external frequency reference to be provided if the internal oscillator is not sufficient.

STM32 Nucleo Board

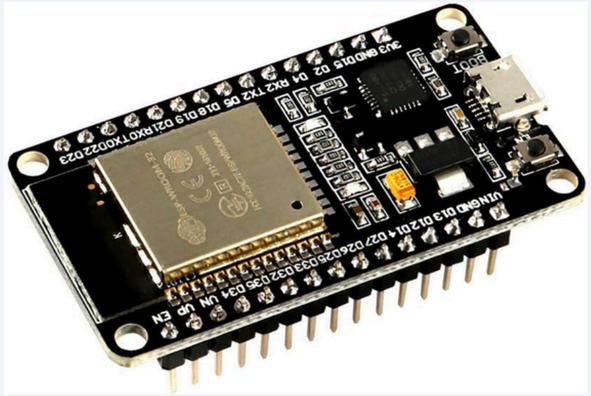


When configuring the project in STM32CubeIDE,we need to use these settings for the SPI1 peripheral to match the MAX30001 requirements:

- Mode: Full-Duplex Master.
- Frame Format: Motorola.
- Data Size: 8 Bits.
- Clock Polarity (CPOL): Low (CPOL = 0).
- Clock Phase (CPHA): 1 Edge (CPHA = 0).
- Baud Rate: The MAX30001 supports up to 20MHz, but for stable breadboard testing, start around 5-10 MHz.

MAX30001 Pin	Nucleo Connector Pin	STM32 GPIO Pin	Function
VCC	5V	5V	Analog Power (5V)
3V3	3V3	3V3	Digital Logic (3.3V)
GND	GND	GND	Common Ground
MISO	D12	**PA6**	SPI1 MISO
MOSI	D11	**PA7**	SPI1 MOSI
SCK	D13	**PA5**	SPI1 SCK
CS0	D10 (or D7)	**PB6**	Hardware/Software CS
INT1	D2	**PA10**	External Interrupt (EXti)

ESP32-DEVKIT (Wireless Module)



- We would establish a dedicated high-speed data path by cross-connecting STM32 UART2 *(TX: PA3/CN10_37, RX: PA2/CN10_35)* to ESP32 UART2 *(RX: GPIO16, TX: GPIO17)*. This configuration isolates the raw medical data stream from the primary USB-to-Serial debug port (UART0 of ESP32).

- We bypass the ESP32’s onboard regulator and dropout losses by connecting the regulated 3.3V rail directly to the 3V3 pin instead of the Vin pin.

PROJECT MANAGEMENT PLAN

TASK BREAKDOWN

JAHNVI

- SPI interfacing of MAX30001s with the STM Nucleo.
- Write ECG R-peak detection and IPG foot detection algorithms
- Interfacing the Nucleo with an ESP32

YASHVARDHAN

- Build the Analog Front End (AFE) on a PCB
- Testing of the components
- Figuring out the current and voltage separation per chip on the two electrodes

KRUNAL

- MAX30001 Testing
- Write ECG R-peak detection and IPG foot detection algorithms
- Create GUI on a laptop to display plots and metrics

SUDHINDRA

- Build the Analog Front End (AFE) on a PCB
- Testing of the components
- Deciding the exact placement (location) of the electrodes

VINAY

- SPI interfacing of MAX30001s with the STM Nucleo.
- Design code architecture
- Design the physical encasing of the device, including user control functionalities

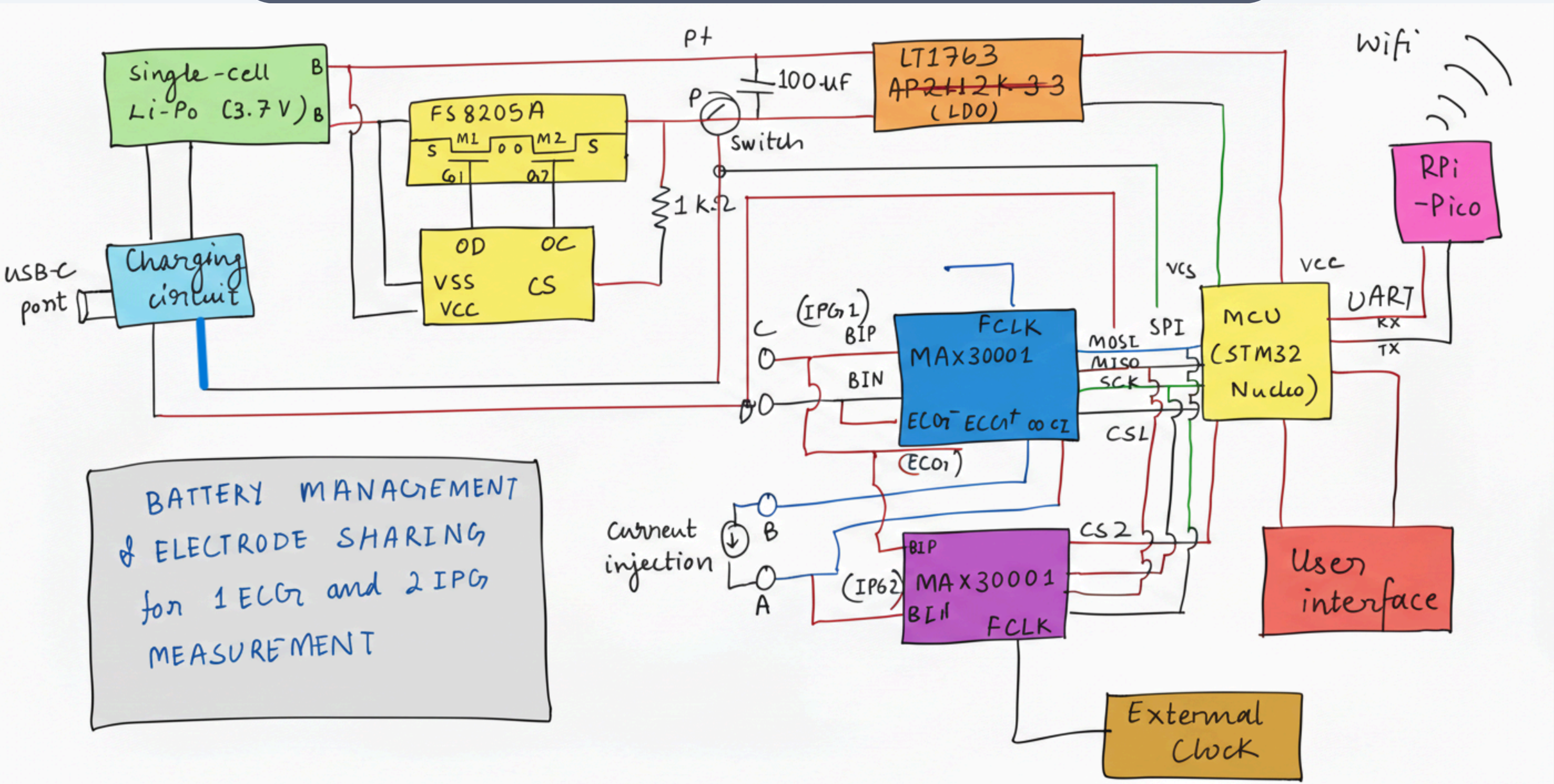
RISKS & ASSUMPTIONS

- Injected Current in IPG may exceed safe or specified limits
- Electrode-skin contact impedance is high and variable, will affect SNR
- AFE input may saturate due to large electrode DC offsets or high common-mode voltage

DOCUMENTATION

- Updated weekly on OneNote, shared on Teams
- Includes details on discussions, their outcomes, project progress details & implementation issues.

BLOCK DIAGRAM

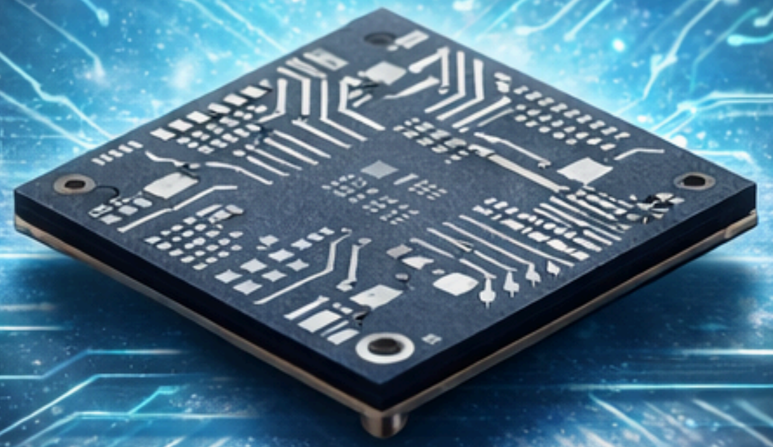


Functionalities of MAX30001

Complete Register-Level Control

- All system parameters are programmmable through SPI-accessible registers. The microcontroller can dynmically configure both ECG and BioZ channels according to application requirements.
- The ECG channel allows adjustment of the sampling rate (125–512 sps), programmable gain (20–160 V/V), digital high-pass filtering (0.5 Hz or bypass), and digital low-pass filtering (40, 100, or 150 Hz).
- The BioZ channel supports configurable sampling rate (25–64 sps), programmable gain (10–80 V/V), digital high-pass and low-pass filtering, selectable current injection magnitude (8–96 µA), and programmable current injection frequency (0.125–1311 KHz).
- Additional configurable parameters include demodulation phase offset, lead bias impedance selection, operating power modes, and internal calibration signal injection.

This register-level flexibility enables precise tuning of signal acquisition, power consumption, and measurement accuracy at runtime.



Functionalities of MAX30001

- **Built-In Signal Intelligence**

The MAX30001 includes integrated signal processing features that reduce the computational burden on the microcontroller. It performs real-time R-to-R detection and supports heart rate calculation directly within the chip. Pacemaker pulse detection is also handled internally, along with configurable window comparison logic for event detection. Beat-to-beat interrupt generation allows the microcontroller to wake only when meaningful physiological events occur, eliminating the need for continuous high-load DSP processing.

- FIFO-Based Data Architecture

Data acquisition is managed through an internal FIFO-based buffering system rather than continuous streaming. The chip provides a 32-word FIFO for ECG data and an 8-word FIFO for BioZ data, enabling efficient burst transfers over SPI. Each sample is delivered in a tagged format that includes status information alongside measurement data. Programmable FIFO thresholds generate interrupts when data is ready, allowing event-driven communication and reducing unnecessary processor activity.

Functionalities of MAX30001

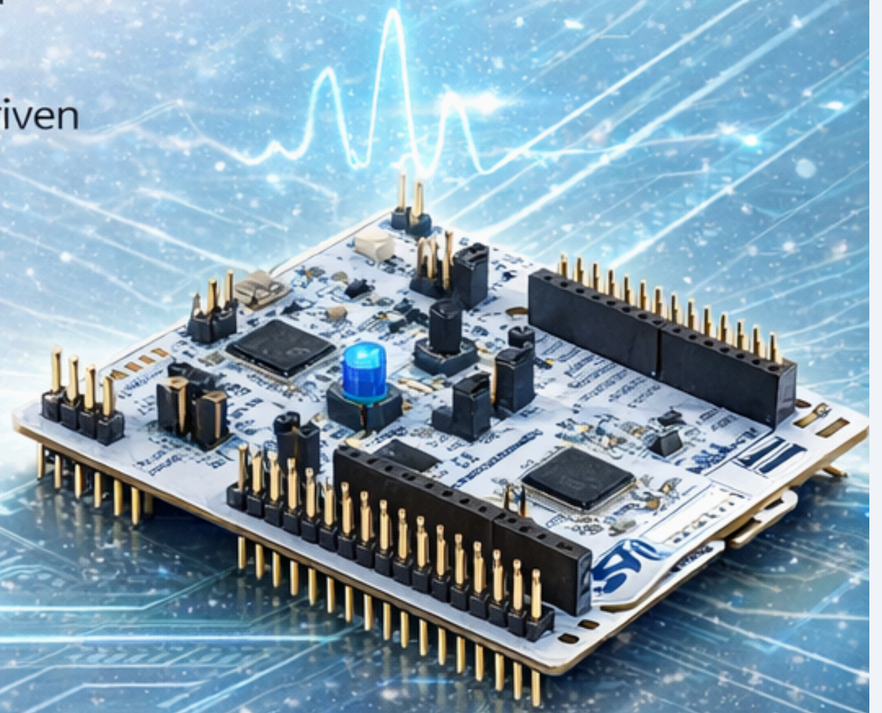
4. Safety and Signal Integrity Controls

- DC lead-off detection injects a very small current (5–100 nA) into the ECG electrodes and detects open or disconnected leads based on voltage deviation.
- For the BioZ channel, AC lead-off detection checks the impedance measurement path and identifies open-circuit conditions during respiration monitoring.
- Lead bias control applies a high-impedance bias toward VMID (50 MΩ, 100 MΩ, or 200 MΩ) to prevent floating inputs without distorting the ECG signal.
- The BioZ driver compliance voltage is defined as:
  VDRVP–VDRVN=±(VAVDD–0.5)V_{DRVN} –v_{DRVN) = \pm (V_{AVDD — 0.5)
  VDRVP–VDRVN≙±(VAVDD–0.5).
- For a 2 V supply, this results in a maximum swing of approximately 1.5 V. With a minimum output impedance of 100 kΩ, the current is limited to approximately 15 µA. The injected current is further constrained by a hardware-limited current DAC and supply-defined compliance limits, ensuring safe operation.

Embedded Processing Unit

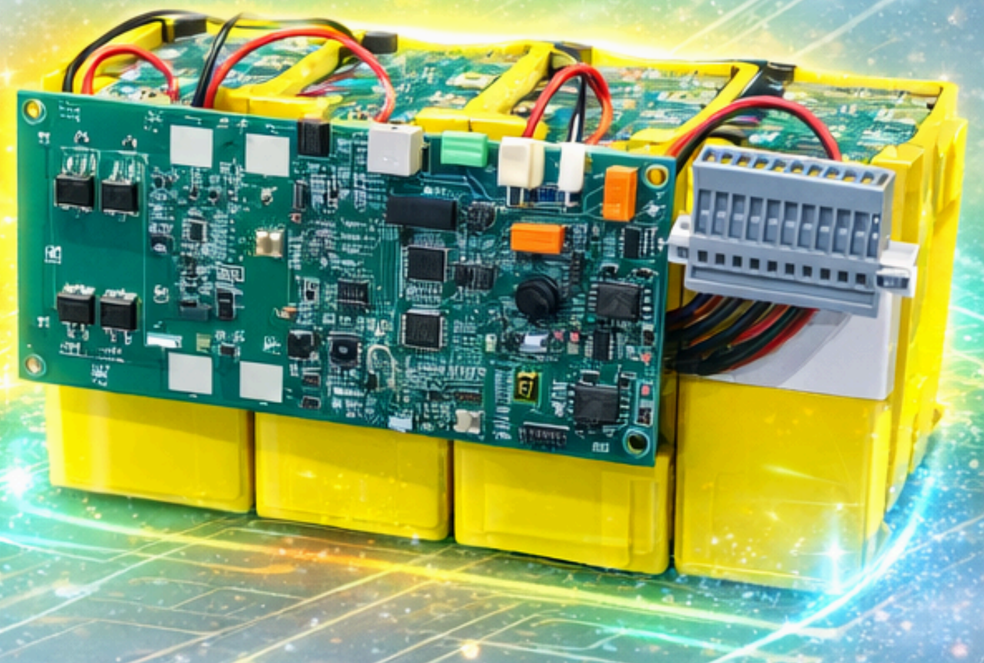
Hardware Interface and Signal Processing Architecture

- The system is built around an STM32 microcontroller operating as the SPI master, interfacing with two MAX30001 devices configured as SPI slaves. Both devices share the SPI bus lines (MOSI, MISO, and SCLK) while using independent chip-select lines for device control. A common 32.768 kHz reference clock is provided to both chips to ensure synchronized sampling, enabling accurate dual-sensor acquisition.
- ECG and dual IPG signals are acquired through the MAX30001 devices and transferred using FIFO-based burst reads over SPI. Data handling is interrupt-driven to reduce processor load and ensure deterministic timing.
- Digital processing is performed on the STM32. R-peaks are detected using a Pan–Tompkins-based approach, while waveform foot detection is applied to the IPG signals. The two IPG waveforms are time-aligned to enable precise delay estimation.
- The R–R interval derived from ECG is used to calculate heart rate, and the foot-to-foot delay between the two IPG signals is used to compute Pulse Transit Time (PTT).



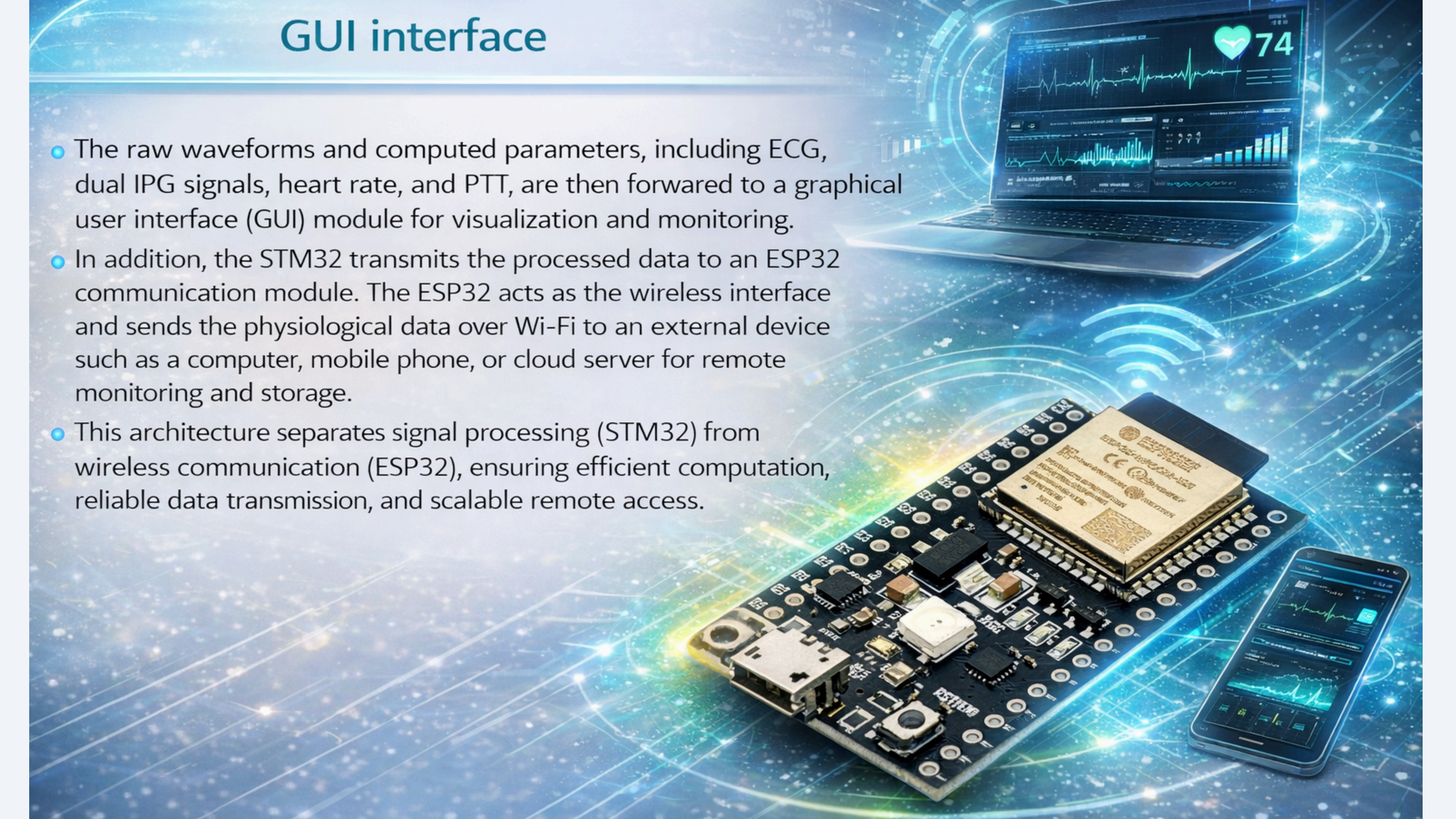
Battery Management and Power Architecture

- The power architecture follows a layered hierarchy: Battery Cell → BMS (DW01A + FS8205A) → Charging / Load Switch → LDO Regulator → System. This structure ensures that protection is applied before charging and power distribution.
- The BMS core, consisting of the DW01A and FS8205A, continuously monitors battery voltage and current. It provides protection against overcharge, over-discharge, overcurrent, and short-circuit conditions. The DW01A controls a dual MOSFET power gate within the FS8205A to disconnect the battery immediately in the event of a fault, forming the the first layer of system safety.
- The TP4056 charging module is connected to the protected P+ and P– terminals and performs controlled constant-current/constant-voltage (CC/CV) charging. If abnormal conditions occur, the BMS can isolate the battery from the charger.
- Load control is managed through the TPS22917, which enables controlled power delivery to the system with soft-start behavior and low leakage characteristics.
- Finally, the MCP1700 low-dropout regulator converts the battery voltage to a stable regulated output (e.g., 3.3 V) for system operation while maintaining low quiescent current consumption.



GUI interface

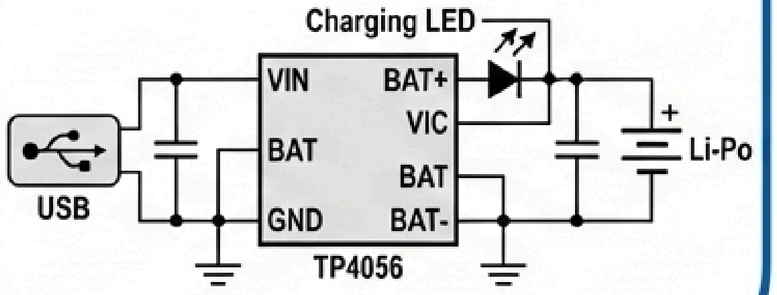
- The raw waveforms and computed parameters, including ECG, dual IPG signals, heart rate, and PTT, are then forwared to a graphical user interface (GUI) module for visualization and monitoring.
- In addition, the STM32 transmits the processed data to an ESP32 communication module. The ESP32 acts as the wireless interface and sends the physiological data over Wi-Fi to an external device such as a computer, mobile phone, or cloud server for remote monitoring and storage.
- This architecture separates signal processing (STM32) from wireless communication (ESP32), ensuring efficient computation, reliable data transmission, and scalable remote access.



PRELIMINARY HARDWARE VALIDATION TESTS

1. BATTERY CHARGING TEST (TP4056 + Li-Po Cell)

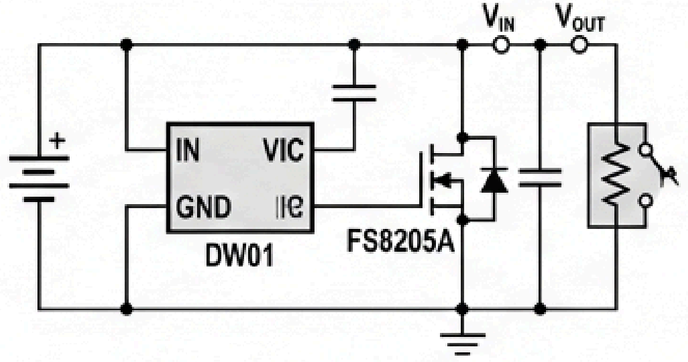
- **Goal:** Verify safe, controlled charging of 3.7V Li-Po battery without overcharging.
- **Method:** Apply 5V USB supply to TP4056 input. Measure battery voltage across BAT+ and BAT- terminals. Observe voltage rise.
- **Pass Criteria:**
  - ✓ Battery voltage rises smoothly.
  - ✓ Charging stops automatically near 4.2V (±0.05V).
  - ✓ No heating of battery or IC.
- **Why:** Ensures safe charging, prevents swelling, explosion, or damage.



TP4056 Charging Circuit

2. PROTECTION CUTOFF TEST (DW01 + FS8205A Protection Circuit)

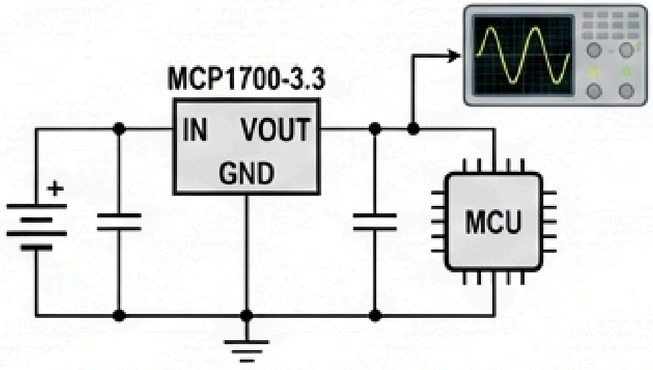
- **Goal:** Verify DW01 & FS8205A disconnect load when battery voltage drops below safe level.
- **Method:** Connect load resistor across system output. Allow slow discharge. Monitor battery and output voltage simultaneously.
- **Pass Criteria:**
  - ✓ Output disconnect occurs near 2.4V – 2.5V.
  - ✓ Output current becomes zero instantly.
  - ✓ Reconnect only after recharge.
- **Why:** Prevents deep discharge, protecting Li-Po battery lifetime.



DW01 Battery Protection Circuit

3. LDO REGULATION TEST (MCP1700 Voltage Regulator)

- **Goal:** Verify MCP1700 LDO provides stable 3.3V output for MCU and sensors.
- **Method:** Power from battery. Measure VIN/VOUT pins with oscilloscope.
- **Pass Criteria:**
  - ✓ Output voltage = 3.3V (±0.05V).
  - ✓ Ripple < 50 mV peak-to-peak.
  - ✓ Stable under load change.
- **Why:** Ensures reliable operation of STM32 and analog front end.



MCP1700 LDO Regulator Circuit

PRELIMINARY TESTING: RESULTS, DISCUSSION, DEVIATIONS, AND IMPROVEMENT PLAN

1. RESULTS SUMMARY

✅ **Battery Charging (TP4056 + Li-Po Cell):** Battery voltage increased smoothly during charging. Charging terminated near 4.2 V, indicating correct CC–CV operation. No abnormal heating observed.

✅ **Protection Circuit (DW01 + FS8205A):** Output disconnected automatically when battery voltage dropped to cutoff region. System resumed operation after reconnecting charger.

✅ **Regulation Stage (MCP1700 LDO):** Output voltage remained stable around 3.3 V. Ripple was minimal and within acceptable limits.

2. DISCUSSION OF PERFORMANCE

- The preliminary tests confirm that all power-management blocks function correctly at the subsystem level.
- The charging IC safely regulates battery voltage, the protection IC prevents unsafe discharge conditions, and the LDO provides a stable supply required for reliable MCU and analog front-end operation.
- Overall, results indicate that the power chain is electrically stable and suitable for further integration testing.

3. DEVIATIONS FROM EXPECTED BEHAVIOUR (IF OBSERVED)

Observation	Possible Cause	Hypothesis
Charging slower than expected	Incorrect charge resistor	Charge current set lower
Cutoff voltage slightly off	Component tolerance	Resistor tolerance or IC threshold variation
Output ripple higher	Capacitor ESR	Output capacitor not optimal

4. HYPOTHESIS-DRIVEN DEBUG STRATEGY

- ✓ If any abnormality appears: Measure individual node voltages to isolate stage causing error
- ✓ Compare measured values with datasheet specifications
- ✓ Replace suspected components with known-good parts
- ✓ Check PCB soldering and grounding integrity

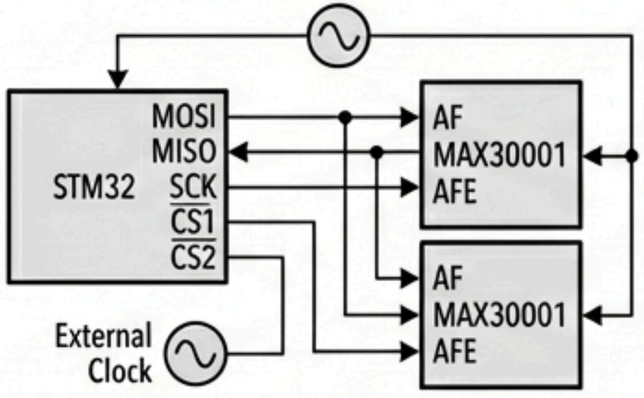
5. PLAN TO ADDRESS OUTSTANDING ISSUES

- If deviations persist, the following corrective actions will be implemented:
  - ➢ Adjust resistor divider values for precise cutoff threshold
  - ➢ Replace capacitors with low-ESR types
  - ➢ Improve PCB grounding and trace routing
  - ➢ Add decoupling capacitors near IC supply pins

INTERMEDIATE TESTS

4. MCU + Sensor Communication Test (STM32 + MAX30001 (SPI Interface))

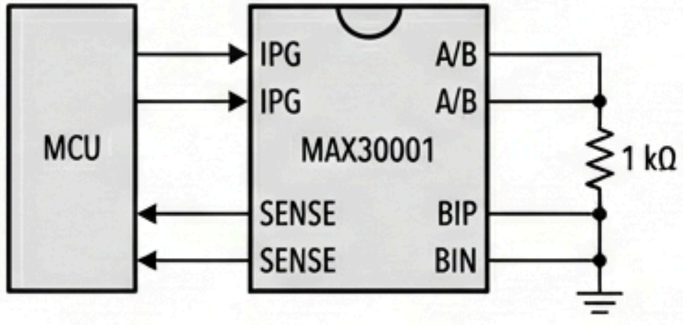
- **Goal:** Verify reliable SPI communication between STM32 MCU and MAX30001 biomedical AFE chips.
- **Method:** Power system. Send SPI register read command from STM32 to MAX30001. Read known register (e.g., device ID). Repeat multiple times.
- **Pass Criteria:**
  - ✓ Correct register value returned every read.
  - ✓ No communication timeout.
  - ✓ No corrupted data bits.
- **Why:** Confirms digital interface integrity before testing analog biosignal paths.



STM32 to Dual MAX30001 SPI Interface

5. Current Injection + Measurement Test (MAX30001 IPG Path)

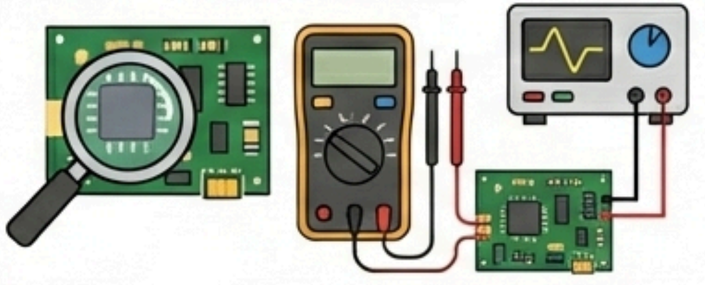
- **Goal:** Verify impedance measurement path and accuracy of current injection + sensing chain.
- **Method:** Replace human body with 1 kΩ precision resistor across electrode inputs. Start IPG measurement mode from MCU. Record measured impedance.
- **Pass Criteria:**
  - ✓ Measured value within ±5% of actual resistor value.
  - ✓ Stable reading over time.
  - ✓ No noise spikes.
- **Why:** Ensures analog front end + firmware + electrodes all work together correctly.



MAX30001 IPG Path Test with Precision Resistor

6. PCB Manufacturing Test (Bare + Assembled Board Validation)

- **Goal:** Verify fabricated PCB has no manufacturing defects before full system testing.
- **Method:** Perform Step-by-Step Checks: Visual Inspection, Continuity Test, Power-On Test, and Rail Verification.
- **Pass Criteria:**
  - ✓ No visible defects, no short circuits (>10 kΩ).
  - ✓ Current draw within expected range, no overheating.
  - ✓ All voltage rails at expected levels.
- **Why:** Detects manufacturing faults early to prevent system damage.



PCB Inspection, Continuity & Power-On Validation

FINAL SYSTEM VALIDATION TESTS

7. Real Signal Acquisition Test — Full System (Electrodes + MAX30001 + STM32 + ESP32)

Goal: Confirm that the complete system correctly captures real biological signals (ECG + dual IPG) from a human subject.

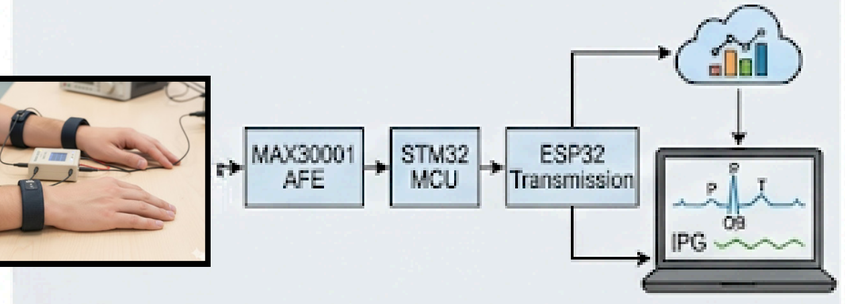
Components Involved:
- Dry electrodes (5-electrode setup)
- MAX30001 Analog Front End ICs
- STM32 microcontroller
- ESP32 transmission module
- MCP1700 regulator + battery stage

Method / Procedure:
- Attach electrodes to subject according to placement diagram.
- Power device using battery.
- Start acquisition firmware.
- Observe waveform on serial plotter / PC dashboard / cloud data.

Expected Behaviour:
- ECG waveform visible with P, QRS, T components.
- IPG signal waveform stable and periodic.

Pass Criteria:
- ✔ Clear QRS peaks visible
- ✔ Stable baseline (minimal drift)
- ✔ No signal clipping or saturation

Why This Test Matters:
Validates full signal chain: electrodes → analog front end → ADC → MCU → transmission.



8. Motion Robustness Test — Wearable Stability Evaluation

Goal: Verify signal reliability under real-world movement conditions.

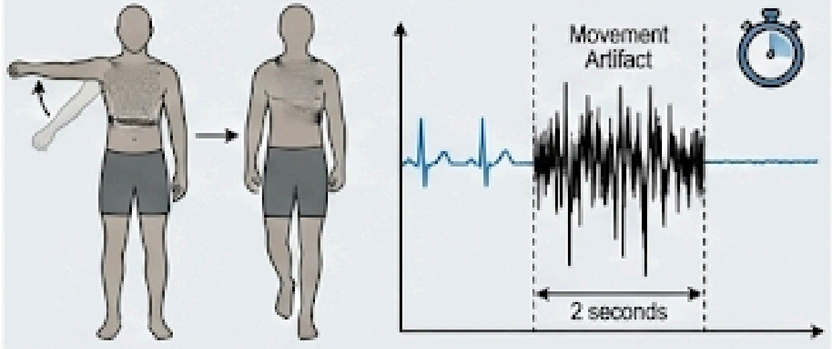
Components Involved:
- Full assembled device
- Electrodes attached to subject
- Data monitoring interface

Method / Procedure:
- Begin recording signals.
- Subject performs movements:
  - Arm motion
  - Walking in place
  - Slight body rotation
- Observe signal quality during and after motion.

Expected Behaviour:
- Temporary noise during movement.
- Signal stabilizes once motion stops.

Pass Criteria:
- ✔ Signal recovers within ≤ 2 seconds
- ✔ No permanent baseline shift
- ✔ No device reset or disconnect

Why This Test Matters:
Wearable biomedical devices must function reliably despite motion artifacts.



9. Endurance / Long-Duration Stability Test — Reliability Test

Goal: Ensure system can operate continuously without failure or unsafe conditions.

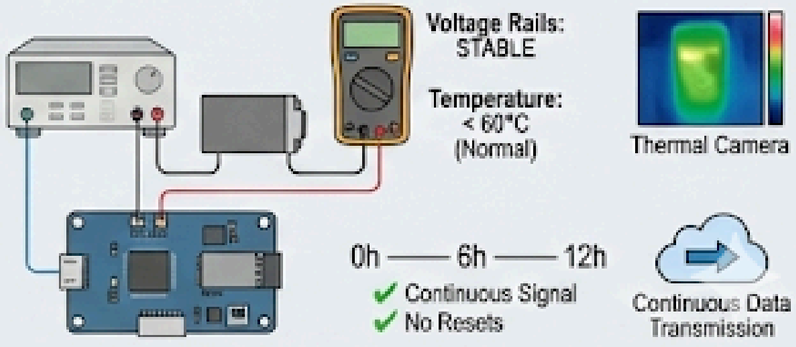
Components Involved:
- Entire device assembly
- Battery supply
- Temperature monitoring (optional thermometer or thermal camera)

Method / Procedure:
- Run device continuously for 6–12 hours.
- Record:
  - Voltage rails
  - Temperature of ICs
  - Signal continuity
- Monitor for resets or communication drops.

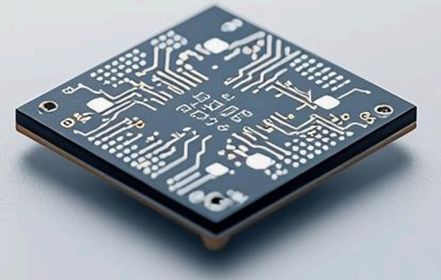
Expected Behaviour:
- Stable voltage rails.
- Constant signal acquisition.
- Normal operating temperature.

Pass Criteria:
- ✔ No overheating (ICs < 60 °C)
- ✔ No MCU reset or firmware crash
- ✔ Voltage rails remain stable
- ✔ Continuous data transmission

Why This Test Matters:
Demonstrates real-world reliability and confirms system is safe for prolonged use.



Thinking ahead to the next milestone



ENGINEERING DESIGN CONSIDERATIONS FOR ASSEMBLING PROTOTYPES

1. POWER INTEGRITY & SAFETY

- Li-Po battery stage must include charging + protection + regulation chain.
- Protection IC + MOSFET must be verified before connecting sensitive electronics.
- Separate analog and digital power traces to prevent noise coupling.
- Ensure LDO output stability with proper input/output capacitors placed close to pins.
- Provide reverse polarity and short-circuit protection during testing.

2. ANALOG SIGNAL INTEGRITY (Critical for ECG/IPG)

- MAX30001 analog front ends require:
  - Short electrode traces
  - Guard rings or ground shielding
  - Low-noise reference routing
- Analog ground must be isolated from digital switching currents.
- External clock trace must be short and shielded to prevent jitter.
- Avoid routing SPI lines parallel to analog inputs.

3. MIXED-SIGNAL PCB LAYOUT STRATEGY

- Partition PCB into zones:
  - Power section
  - Analog front end section
  - Digital MCU section, RF/WiFi section.
- Use star grounding scheme.
- Place decoupling capacitors near every IC supply pin.
- Keep crystal/clock lines away from switching regulators and WiFi traces.

4. ELECTRODE INTERFACE RELIABILITY

- Provide ESD protection at electrode inputs.
- Use medical-grade connectors or gold-plated pads.
- Include test pads for signal probing during debugging.

5. FIRMWARE–HARDWARE CO-DESIGN

- SPI bus must support two MAX30001 devices via separate chip select lines.
- Include test firmware modes: register read test, dummy signal injection mode.
- Provide UART debug interface for diagnostics.

6. MECHANICAL & ASSEMBLY CONSIDERATIONS

- Battery placementably front end.

7. DEBUG & TESTING PROVISIONS

- Include test points for: battery voltage, 3.3 V rail, SPI lines, electrode inputs.
- Add jumper pads to isolate subsystems during debugging.

VENDOR IDENTIFICATION & OUTSOURCING PLAN

Vendor Selection

- Based on PCB design requirements
- Trace width, spacing, pad pitch
- Fine-pitch ICs tolerances

Primary Preference

- Institute Facility: IITB PCB Lab
- Faster iteration, comms, lower cost
- Rapid design modifications

Secondary Vendors (if higher precision needed)

- External Professional Vendors:
  - PCB Power
  - Lion Circuits
- Capability, Turnaround Time, Cost

Decision Criteria

Minimum Trace/Space Capability

↓

Multilayer Support

↓

Manufacturing Accuracy

↓

Delivery Time

↓

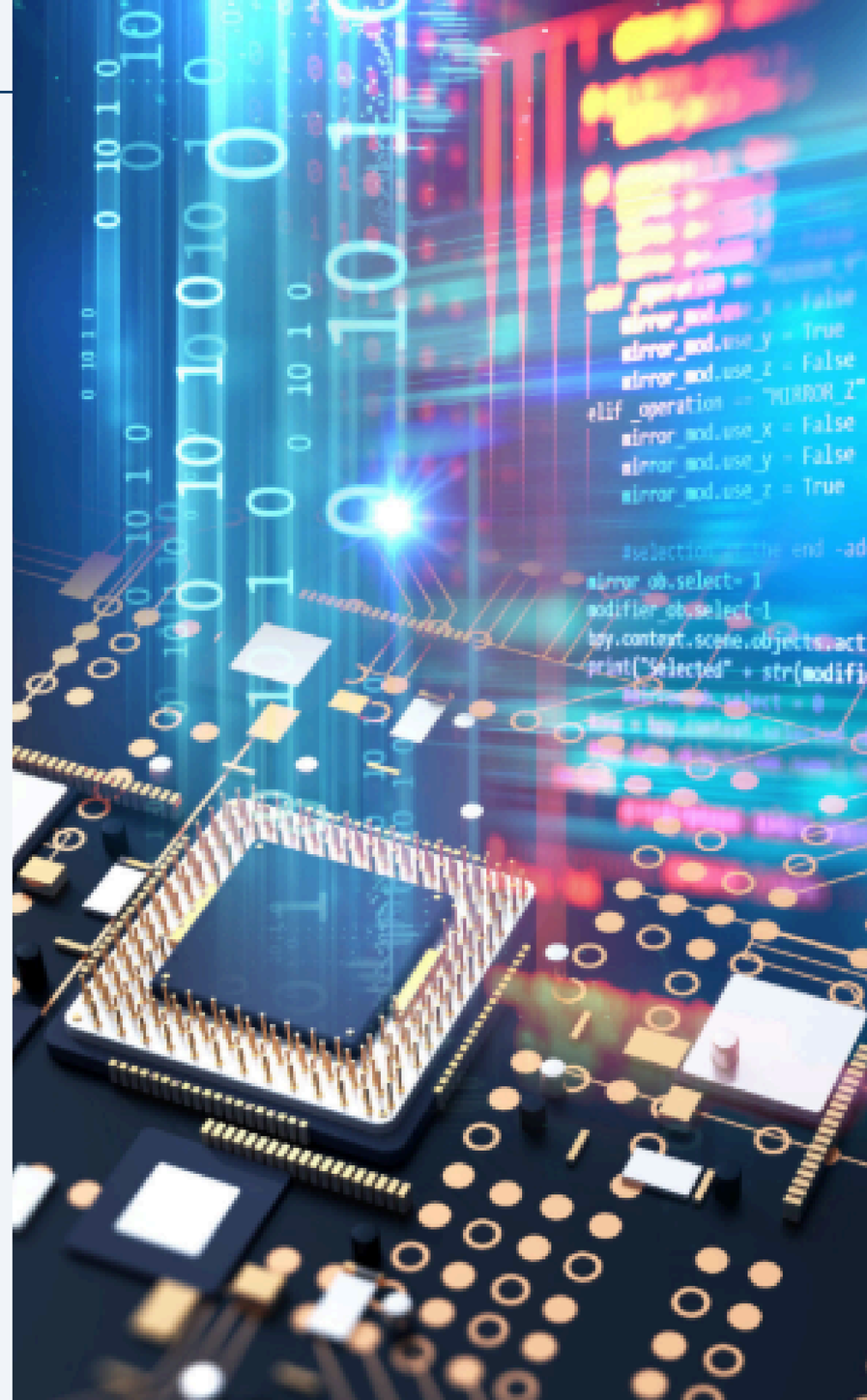
Budget Constraints

Vendor selection finalized after PCB design rule analysis for manufacturability and reliability

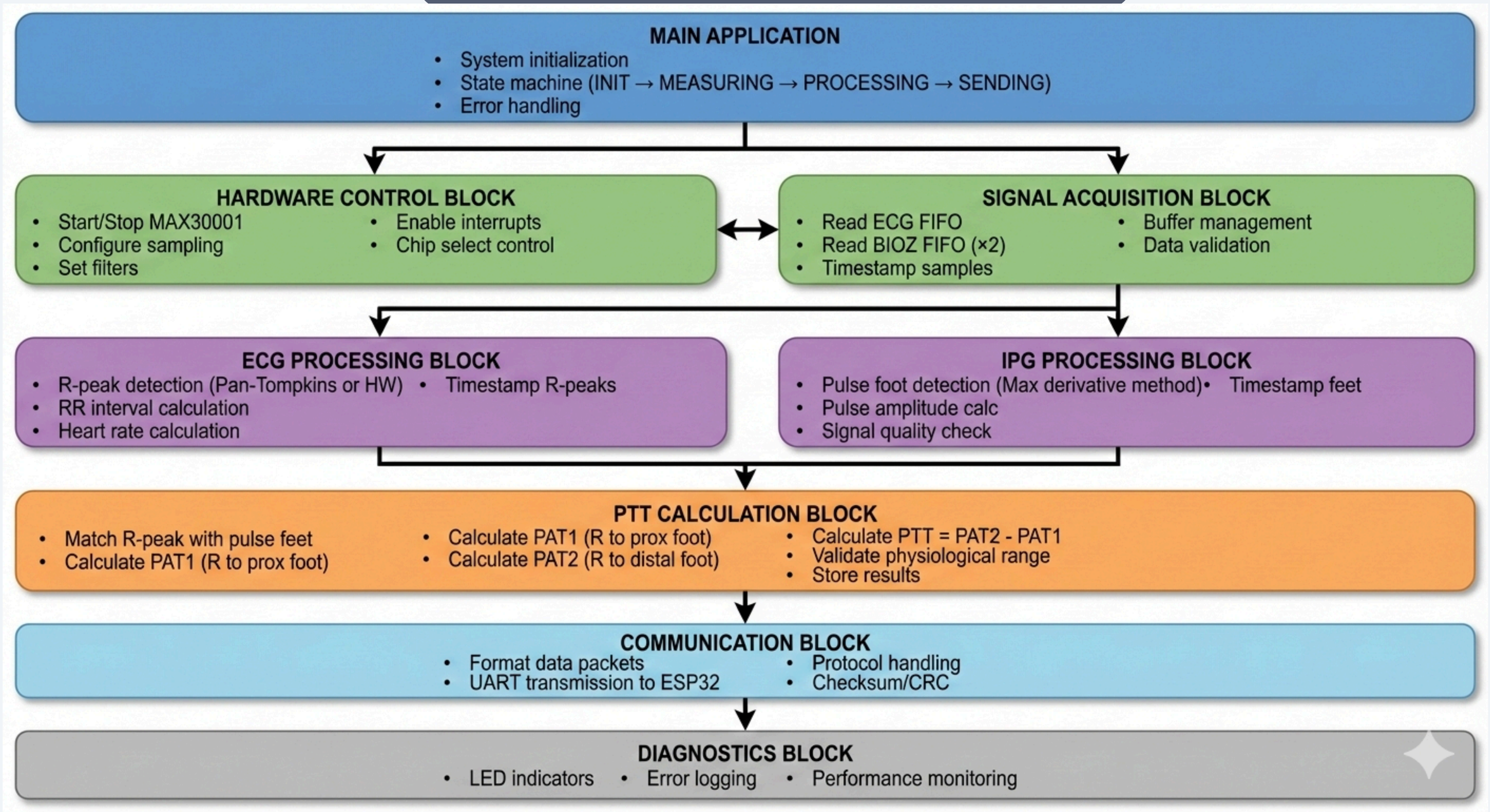
System Software

Some Design Decisions and Rationale:

- RTOS: ensures deterministic sampling (ECG 256 Hz, IPG 128 Hz), ms-accurate timestamps for PTT, and priority-based execution so acquisition and processing always preempt communication and UI tasks.
- RTOS Choice (vs Zephyr): selected FreeRTOS for lower RAM/Flash usage, simpler STM32 integration, and sufficient real-time features without unnecessary complexity.
- No DMA: data transfers are small and interrupt-based drivers reduce system complexity.
- Logic Analyzer for Debugging: enables verification of SPI timing, chip-select sequencing, and interrupt behavior that cannot be reliably observed via software debugging alone.

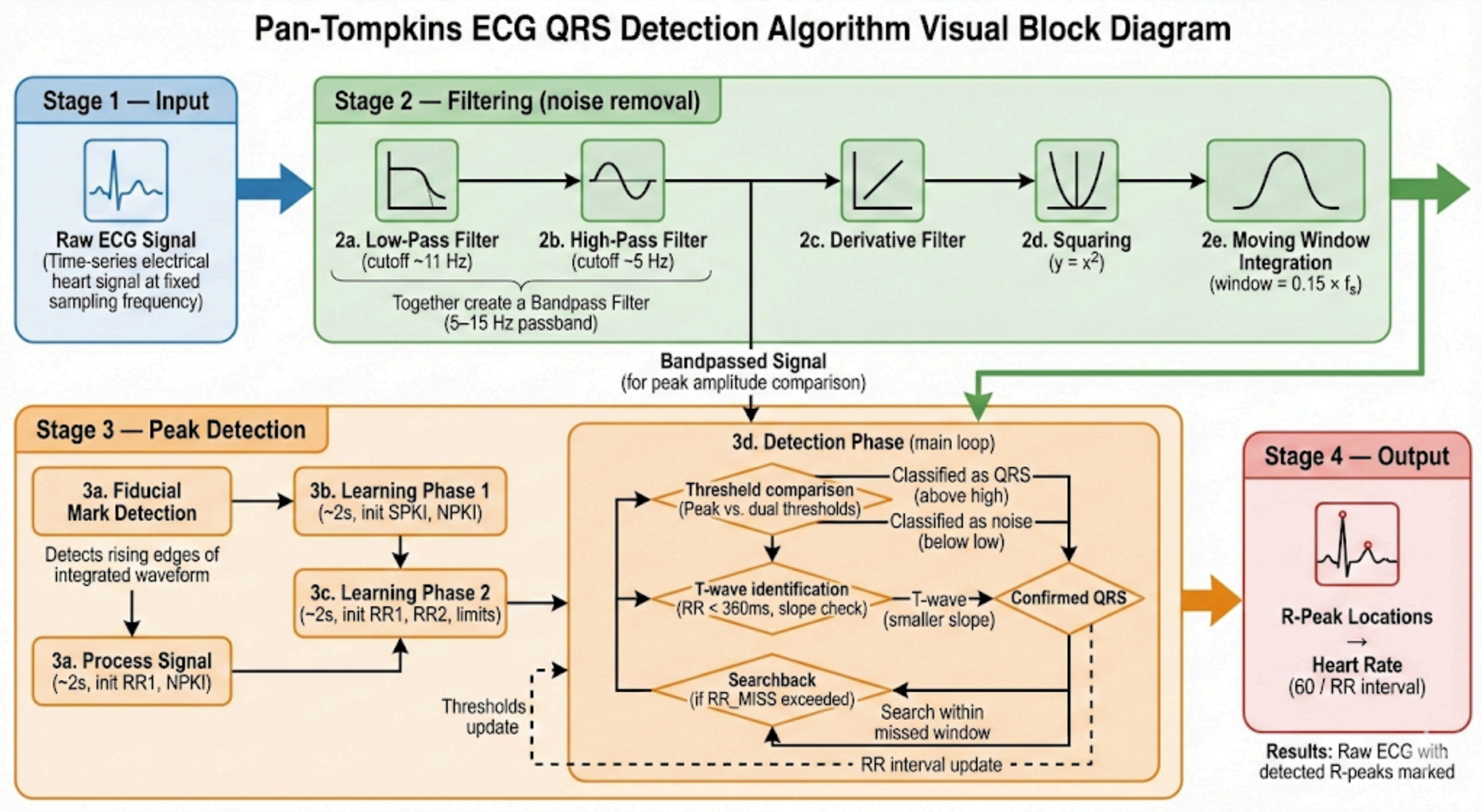


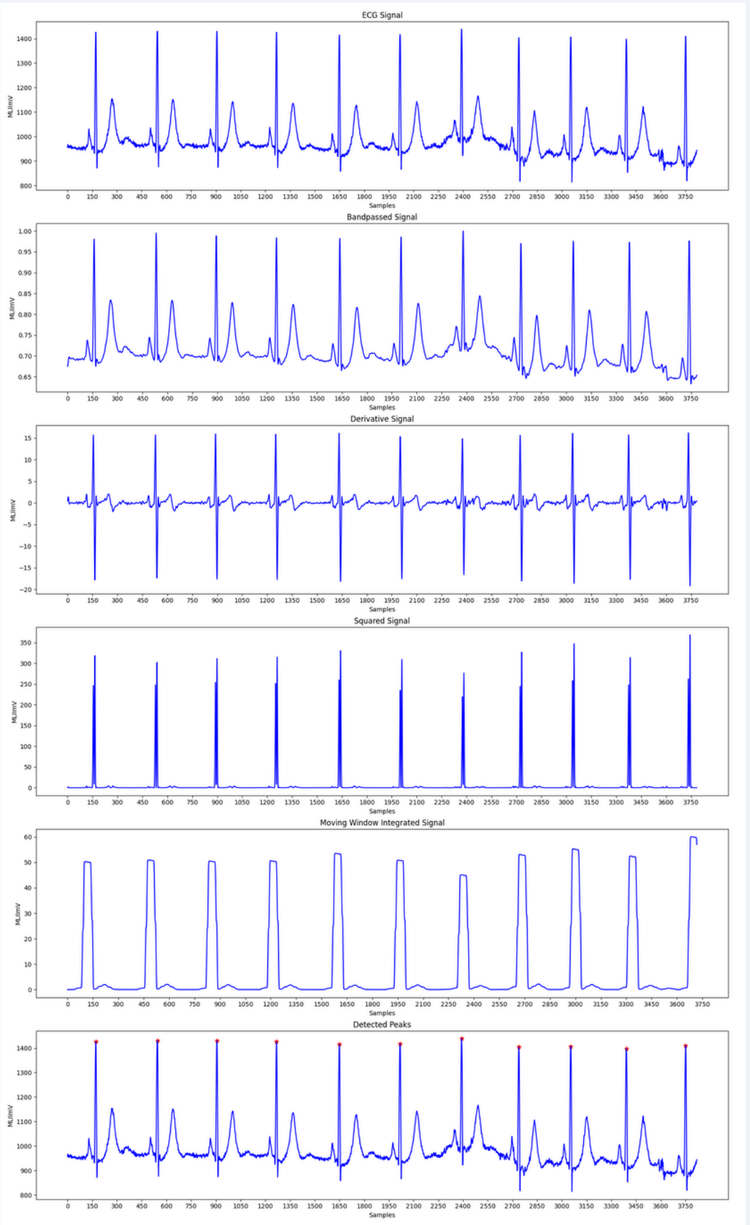
Code Architecture



Simulation: R-peak detection algorithm

The algorithm is a linear signal processing pipeline in which each stage transforms the ECG signal and passes it to the next. The final stage uses outputs from two stages simultaneously (filtered signal + integrated signal) for decision-making. Another one for IPG processing will also be written.





Non-Electrical Design Considerations

1' Electrode Selection Strategy

Wet (Gel-Based) Electrodes

- Lower Ionic Conduction
- Reduced Motion Artifact

Dry Electrodes

- Reduced Motion Artifact
- Stable Baseline

2' Electrode–Skin Interface Impedance

Z_α (kΩ) kiloohms to hundreds kΩ)

- Depends on:
- Material on: Material, Area, Pressure,
- Moisture, Pre, Skin Prep, Gel Quality

- Higher Z_α leads to: Noise, Interference, Poor CMRR

3' Material Considerations

Ag/CCl Electrodes

- Stable Potential
- Biocompatible

Mechanical Housing

- Firm Contact
- User Comfort

4' Thermal and Environmental Factors

Z_α Varies with:

- Moisture, Temp, Sweat, Movement,
- Moduile, Location

Testing Conditions:

- Posture, Posture, Acute Activity, Cable Strain, Contact Pressure

Consistent Environmental & Physiological Conditions = Reliable Measurements